

ECON6102 Macroeconomics for Honours Course Notes

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ECON6102 is a highly comprehensive economics course. It is macroeconomics dressed in the language of mathematics and programming. This also reflects the direction in which modern economics is moving: using mathematical reasoning and computational methods to formulate, analyse, and explain economic questions.

These notes are among the ones I am most satisfied with so far. They are well structured and relatively complete in coverage. I used the course materials from Semester 1, 2025 to compile most of the notes, and then refined them using the 2026 materials. Of course, Andres's lecture notes are already very well developed; my contribution here is mainly to organise and bind them into a coherent set of notes. If there is one limitation, it is probably that I have not included the Matlab content.

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Part I

Dynamic General Equilibrium Models in Deterministic Frameworks

1 Ordinary Differential Equations

Dynamic optimisation in continuous time typically involves analysing Ordinary Differential Equations (ODEs).

Unlike “algebraic” equations, in “differential” equations:

- The unknown is a function, not a variable.
- The equation includes one or more of the derivatives of the function.

An “ordinary” differential equation is one for which the unknown is a function of only one variable. In “Partial” differential equations, on the other hand, the unknown is a function of more than one variable.

1.1 Discrete and Continuous Time

Dynamic macroeconomic models can either be formulated in discrete time, or in continuous time.

- In discrete time, periods are indexed by $t = 0, 1, 2, \dots, T$, with $T \leq \infty$.
 - More generally, intervals between decision periods are discrete.
- In continuous time, periods are indexed by $t \in [0, T]$.
 - Intervals between decision periods is arbitrarily small.

Continuous time models are essentially “the limit” of discrete time models, as intervals between periods tend to zero.

Example 1.1.1. Law of motion for capital per capita $k(t)$ in discrete-time Solow model:

$$k(t+1) = sf(k(t)) + (1-\delta)k(t)$$

where $s \in (0, 1)$ is the savings rate, $\delta \in (0, 1)$ is the depreciation rate and f is the production function (I assume zero population growth).

Question: How does $k(t)$ change in only a fraction of the time between periods t and $t+1$?

Answer: Let $\Delta \in (0, 1)$, then:

$$k(t+\Delta) = \Delta sf(k(t)) + (1-\delta\Delta)k(t)$$

Rearranging:

$$\frac{k(t + \Delta) - k(t)}{\Delta} = sf(k(t)) - \delta k(t)$$

This holds for any Δ , in particular:

$$\lim_{\Delta \rightarrow 0} \frac{k(t + \Delta) - k(t)}{\Delta} = sf(k(t)) - \delta k(t)$$

Or

$$\dot{k}(t) \equiv \frac{dk(t)}{dt} = sf(k(t)) - \delta k(t)$$

This is the law of motion for capital per capita in the **continuous time version** of the Solow model.

Both discrete and continuous time models have advantages and caveats. Discrete time models can be more intuitive to work with (agents make decisions in discrete time!) + algebra is more straightforward but in continuous time explicit closed-form solutions are more common + numerical computation can be much faster.

In the end, it all boils down to the application:

- Much of growth theory using deterministic frameworks relies on continuous time models.
- Macro frameworks with uncertainty (e.g., heterogeneous agents, or real business cycle theory) typically use discrete time models.

In this class, we will use both methods.

- Part I (Growth) $\rightarrow$ continuous time. We concentrate on deterministic problems. Stochastic problems in continuous time are outside of our scope.
- Part II (Uncertainty) $\rightarrow$ discrete time

1.2 First-Order ODEs

A first-order ODE has the form:

$$\dot{x}(t) = F(t, x(t)) \tag{i}$$

where $\dot{x}(t) \equiv dx(t)/dt$, and $t \in [0, T]$.

Note:

- Equation (i) is a first-order ODE because it only involves the first derivative of $x(t)$.
- The unknown in (i) is a function $x(t)$, with $x : [0, T] \rightarrow \mathbb{R}$.

Is this solution unique? No: (i) admits infinitely many solutions indexed by an integrating constant, say C . Generally, the constant C can be uniquely determined by requiring the solution to pass through a given point on the tx -plane.

Example 1.2.1.

$$\dot{x}(t) = 2t \text{ with } t \in [0, \infty).$$

General solution is:

$$x(t) = t^2 + C$$

Now suppose we impose the initial condition: $x(0) = 1$. Then $C = 1$ and the particular solution is:

$$x(t) = t^2 + 1$$

In this example, we pinned down the unknown constant C via the initial condition $x(0) = 1$. More generally, an ODE is specified together with either an initial condition or with a boundary condition:

- Initial conditions specify the value of $x(t)$ at $t = 0$:

$$x(0) = x_0$$

- Boundary conditions specify the value of $x(t)$ at $t_b \in (0, T]$:

$$x(t_b) = x_b$$

In the previous example we could solve the ODE explicitly/analytically. This can only be done in special cases, depending on the functional form of $F(t, x)$.

A particular type of ODE with closed-form solution is of central importance in dynamic macroeconomic analysis:

“Linear First-Order ODEs”

Two clarifications before moving on:

1. We concentrate on “first-order” ODEs throughout the course. In fact, this is without loss of generality (more on this in a bit...). So from now on, I only leave it implicit that ODEs are “first-order.”
2. From now on, we will mostly concentrate on infinite horizon models, i.e.,

$$T = \infty$$

unless stated otherwise.

Definition 1.2.2. A linear ODE takes the general form:

$$\dot{x}(t) = a(t)x(t) + b(t)$$

with $t \in [0, \infty)$.

In this class, we will mostly focus on cases in which

$$a(t) = a \quad \text{and} \quad b(t) = 0$$

which leads to

$$\dot{x}(t) = ax(t)$$

This type of equation is called a **linear first-order homogeneous ODE with constant coefficients**.

Lemma 1.2.3. Consider the linear ODE:

$$\dot{x}(t) = ax(t) \quad (\star)$$

with $a \neq 0$, and initial condition $x(0) = x_0$. Then the particular solution to $(\star)$ is given by

$$x(t) = x_0 \exp(at)$$

Method 1: Guess and verify

Proof. The “claimed” solution is:

$$x(t) = x_0 \exp(at) \quad (1)$$

Then:

$$\dot{x}(t) = x_0 \exp(at) a = a x_0 \exp(at) = a x(t) \quad \text{by (1)} \Rightarrow \dot{x}(t) = ax(t)$$

□

Method 2: “Integrating Factor method”

Proof. Recall:

$$\dot{x}(t) = ax(t) \Leftrightarrow \dot{x}(t) - ax(t) = 0$$

Suppose I multiply both sides by $\exp(-at)$:

$$\exp(-at) [\dot{x}(t) - ax(t)] = 0. \quad (2)$$

It is not easy to show

$$\frac{d}{dt} (\exp(-at)x(t)) = (-a) \exp(-at)x(t) + \exp(-at)\dot{x}(t) \quad (3)$$

Then, using (3) into (2) (after integrating):

$$\exp(-at)x(t) = -C \Rightarrow x(t) = -C \exp(at)$$

To recover the particular solution, we apply the initial condition $x(0) = x_0$:

$$x(0) = x_0 \Leftrightarrow x_0 = -C$$

Hence:

$$x(t) = x_0 \exp(at)$$

□

Definition 1.2.4. A steady-state of a differential equation in $x(t)$ is a value of $x(t)$ such that

$$\dot{x}(t) = 0 \quad \text{for all } t$$

From now on, we denote a steady state using an asterisk, e.g., x^* . Think of x^* as a “rest point”: once I reach it, I stay there forever. Formally, if $x(t_a) = x^*$, for some t_a , then $x(t) = x^*$ for all $t \geq t_a$.

Example 1.2.5. The steady state of $(\star)$ is $x^* = 0$. In this example, the steady state x^* is unique, but that is not always the case!

Suppose we start from an arbitrary $x(0)$. Would $x(t)$ eventually reach/approach a steady state x^* ? The answer to this question is informative on the dynamics of $x(t)$, i.e., on how it behaves over the short- and long-run. Formally, we are interested in assessing “stability.”

Definition 1.2.6. The steady state $x^* = 0$ is stable, if $x(t) \rightarrow x^*$ for any initial condition $x(0)$. Otherwise, the steady state x^* is unstable.

Notation: $x(t) \rightarrow x^*$ is the same as $\lim_{t \rightarrow \infty} x(t) = x^*$.

Example 1.2.7. Consider the linear ODE

$$\dot{x}(t) = ax(t), \quad \text{with } a \neq 0.$$

Is the steady state $x^* = 0$ stable?

Answer: It depends on the value of a ! See Figure 1.

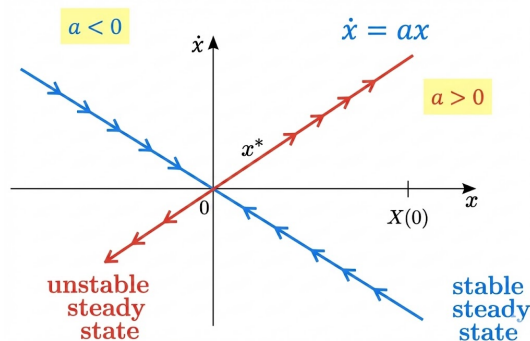


Figure 1: Graphical analysis I of Example 1.2.7

Recall $x(t) = x_0 \exp(at)$. Suppose $x_0 > 0 = x^*$. Then we can plot $x(t)$ against t as:

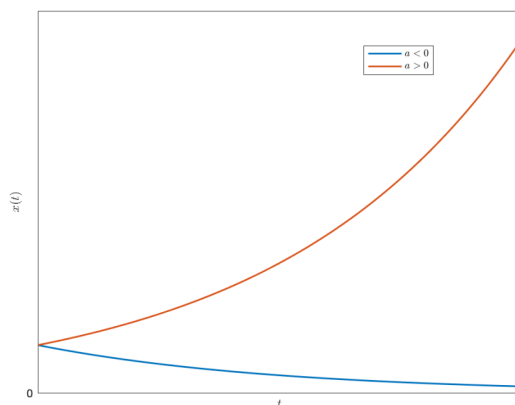


Figure 2: Graphical analysis II of Example 1.2.7

Suppose $x_0 < 0 = x^*$. Then we can plot $x(t)$ against t as:

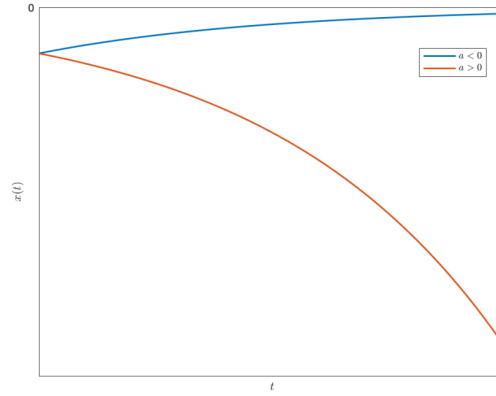


Figure 3: Graphical analysis II of Example 1.2.7

Lemma 1.2.8. *Consider the linear ODE:*

$$\dot{x}(t) = ax(t) \quad (\star)$$

with $a \neq 0$, and initial condition $x(0) = x_0$. Then the unique steady state $x^ = 0$ is stable if and only if $a < 0$. Otherwise, x^* is unstable.*

1.3 Systems of ODEs

A system of ODEs is a set of two or more ODEs, involving two or more variables/unknowns. In this course, we focus on systems of two equations in two unknowns (i.e., 2×2 systems):

$$\dot{x}_1(t) = F_1(t, x_1(t), x_2(t))$$

$$\dot{x}_2(t) = F_2(t, x_1(t), x_2(t))$$

where $t \in [0, T]$.

The general solution typically depends on two arbitrary constants, say C_1 and C_2 , and can then be written as:

$$x_1(t) = \varphi_1(t; C_1, C_2), \quad x_2(t) = \varphi_2(t; C_1, C_2).$$

C_1 and C_2 can be pinned down via two conditions on the solution.

Two types of problems emerge, depending on the nature of such conditions:

1. An Initial Value Problem (IVP):

$$x_1(0) = x_{10} \quad x_2(0) = x_{20}$$

2. A Boundary Value Problem (BVP):

e.g.

$$x_1(0) = x_{10} \quad x_2(T) = x_{2T}$$

Important: Many other boundary conditions would also determine C_1 and C_2 !

As in single-equation ODEs, exact solutions to ODE systems can only be obtained under special cases.

In what follows, we concentrate on 2×2 systems of the form:

$$\dot{x}_1(t) = a_{11}x_1(t) + a_{12}x_2(t)$$

$$\dot{x}_2(t) = a_{21}x_1(t) + a_{22}x_2(t)$$

This is a homogeneous system of linear ODEs with constant coefficients. We will show that the main results of the analysis that follows can be generalised to other linear systems of interest.

We can write the previous 2×2 linear system in matrix form as:

$$\underbrace{\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix}}_{\dot{\mathbf{x}}(t)} = \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}}_{\mathbf{x}(t)} \iff \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

Our goal is to analyse solutions to $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ analytically, and graphically. We proceed in two steps:

1. Diagonal Systems
2. Nondiagonal Systems

1.3.1 Diagonal Systems

Let's start with a special case in which $a_{12} = a_{21} = 0$. In this case, $\mathbf{A}$ is a diagonal matrix:

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix}$$

so that the system boils down to

$$\dot{x}_1(t) = a_{11}x_1(t)$$

$$\dot{x}_2(t) = a_{22}x_2(t)$$

The system is actually “decoupled” in the sense that there is no direct interaction between x_1 and x_2 . The steady state is $\mathbf{x}^* = \mathbf{0}$.

Each equation in a diagonal system is a linear ODE. So based on our earlier analysis, the (general) solution is:

$$x_1(t) = C_1 \exp(a_{11}t)$$

$$x_2(t) = C_2 \exp(a_{22}t)$$

Using this analytical solution, the dynamics of the system depend on the signs of a_{11} and a_{22} :

1. If $a_{11} < 0$ and $a_{22} < 0$, the steady state $\mathbf{x}^*$ is stable. Starting from any $\mathbf{x}(0)$, we have $\mathbf{x}(t) \rightarrow \mathbf{x}^*$.
2. If $a_{11} > 0$ and $a_{22} > 0$, the steady state $\mathbf{x}^*$ is unstable. Starting from any $\mathbf{x}(0)$, $\mathbf{x}(t)$ diverges away from the steady state.
3. What if a_{11} and a_{22} have opposite signs?

In this case, we say that the steady state is saddle-path stable. Essentially, the system converges to the steady state along one dimension, but diverges along the other dimension.

Example 1.3.1. Suppose $a_{11} < 0$ and $a_{22} > 0$. Recall that the solution is:

$$x_1(t) = C_1 \exp(a_{11}t) \quad x_2(t) = C_2 \exp(a_{22}t).$$

The system pushes towards the steady state along the x_1 axis, but it pushes away from the steady state along the x_2 axis.

Important: In cases of saddle-path stability, we can show that there is a unique trajectory going through the steady state which is stable.

Definition 1.3.2. It is convenient to study the dynamics of 2×2 ODE systems by means of **Phase Diagrams**. This is a graphical analysis which can be applied to any type of 2×2 ODE (linear, nonlinear, etc.). It is particularly useful when systems do not admit closed-form solutions.

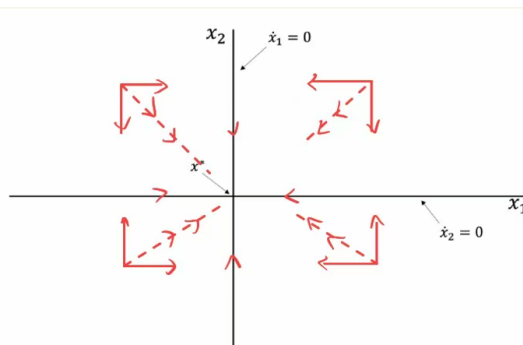
Consider the diagonal system:

$$\dot{x}_1(t) = a_{11}x_1(t)$$

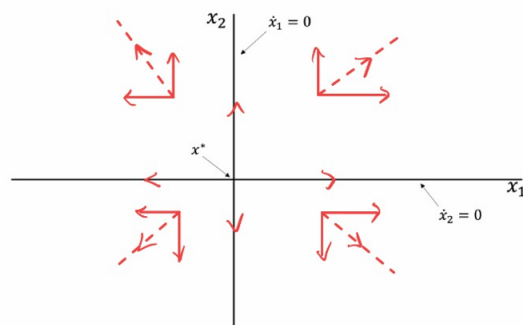
$$\dot{x}_2(t) = a_{22}x_2(t)$$

The phase diagram associated with this system is a plot in (x_1, x_2) , which illustrates the dynamics of the variables using arrows.

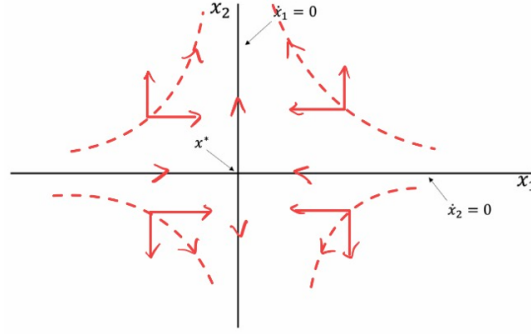
1. Stable steady state ($a_{11} < 0, a_{22} < 0$) :



2. Unstable steady state ($a_{11} > 0, a_{22} > 0$) :



3. Saddle-path stability ($a_{11} < 0, a_{22} > 0$) :



Important: Two-dimensional ODE systems which exhibit saddle-path stability have only two paths that go through the steady state:

1. The “stable arm” (which in the previous graph is the horizontal axis). The stable arm is also the unique stable trajectory in the system.
2. The “unstable arm” (which in the previous graph is the vertical axis).

1.3.2 Nondiagonal Systems

Our next goal is to characterise the solution of the 2×2 ODE system:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

but in the more general case in which $\mathbf{A}$ is nondiagonal, i.e., $a_{12} \neq 0$ or $a_{21} \neq 0$.

At first, we cannot apply the solution methods studied so far. But the idea is to make a change of variables $\mathbf{x}(t) \mapsto \mathbf{y}(t)$ such that we can write the system as

$$\dot{\mathbf{y}}(t) = \mathbf{D}\mathbf{y}(t)$$

where $\mathbf{D}$ is a diagonal matrix.

This can be done by applying Eigenvalues and Eigenvectors.

Definition 1.3.3. Let $\mathbf{A}$ be an $n \times n$ matrix. An eigenvalue of $\mathbf{A}$ is a scalar $\alpha \in \mathbb{R}$ such that

$$\mathbf{A}\mathbf{v}_\alpha = \alpha\mathbf{v}_\alpha \tag{1}$$

for some $n \times 1$ nonzero vector $\mathbf{v}_\alpha$.

Definition 1.3.4. The vector $\mathbf{v}_\alpha$ in equation (1) is an eigenvector of $\mathbf{A}$ associated with the eigenvalue α .

How to compute eigenvalues and eigenvectors?

We start by rearranging equation (1) as

$$(\mathbf{A} - \alpha\mathbf{I})\mathbf{v}_\alpha = \mathbf{0} \tag{2}$$

Suppose $(\mathbf{A} - \alpha\mathbf{I})$ is invertible. Then (2) gives

$$\mathbf{v}_\alpha = (\mathbf{A} - \alpha\mathbf{I})^{-1}\mathbf{0} = \mathbf{0}$$

But we are looking for the solutions to (1) such that $\mathbf{v}_\alpha \neq \mathbf{0}$! Hence, if we want solutions to (1) with $\mathbf{v}_\alpha \neq \mathbf{0}$, then $(\mathbf{A} - \alpha \mathbf{I})$ should be noninvertible.

Equivalently, what we need is that

$$\det(\mathbf{A} - \alpha \mathbf{I}) = 0 \quad (3)$$

Equation (3) is called the characteristic equation of matrix $\mathbf{A}$.

- This is an n -th degree polynomial equation in α with n solutions.
- Each of the solutions to (3), say $\{\alpha_i\}_{i=1}^n$, is an eigenvalue of $\mathbf{A}$.
- Using α_i back into the system (2) we can recover the eigenvector $\mathbf{v}_{\alpha_i}$.

Example 1.3.5. Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}$$

1. Show that the eigenvalues of $\mathbf{A}$ are $\alpha_1 = 3$ and $\alpha_2 = -2$.
2. Show that eigenvectors satisfy:

$$\mathbf{v}_{\alpha_1} = \begin{bmatrix} 2k \\ k \end{bmatrix}, \quad \mathbf{v}_{\alpha_2} = \begin{bmatrix} -\frac{1}{2}k \\ k \end{bmatrix}$$

where k is an arbitrary constant.

Answer.

1. Characteristic equation:

$$\det(\mathbf{A} - \alpha \mathbf{I}) = 0 \quad \Rightarrow \quad \det \left(\begin{bmatrix} 2 - \alpha & 2 \\ 2 & -1 - \alpha \end{bmatrix} \right) = (2 - \alpha)(-1 - \alpha) - 4 = 0$$

This quadratic has two different roots:

$$\alpha_1 = 3, \quad \alpha_2 = -2$$

2. Eigenvectors satisfy:

$$\mathbf{A} \mathbf{v}_{\alpha_i} = \alpha_i \mathbf{v}_{\alpha_i}, \quad i \in \{1, 2\}.$$

Then

$$(\mathbf{A} - \alpha_i \mathbf{I}) \mathbf{v}_{\alpha_i} = \mathbf{0} \quad \Rightarrow \quad \begin{bmatrix} 2 - \alpha_i & 2 \\ 2 & -1 - \alpha_i \end{bmatrix} \begin{bmatrix} v_{\alpha_i}^1 \\ v_{\alpha_i}^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Expanding gives us

$$(2 - \alpha_i)v_{\alpha_i}^1 + 2v_{\alpha_i}^2 = 0$$

$$2v_{\alpha_i}^1 - (1 + \alpha_i)v_{\alpha_i}^2 = 0$$

Consider the first eigenvalue $\alpha_1 = 3$.

Using $\alpha_1 = 3$ into the system above we get:

$$-v_{\alpha_1}^1 + 2v_{\alpha_1}^2 = 0 \quad \Rightarrow \quad v_{\alpha_1}^1 = 2v_{\alpha_1}^2$$

$$2v_{\alpha_1}^1 - 4v_{\alpha_1}^2 = 0 \quad \Rightarrow \quad v_{\alpha_1}^1 = 2v_{\alpha_1}^2$$

Then:

$$\mathbf{v}_{\alpha_1} = \begin{bmatrix} 2k \\ k \end{bmatrix} \quad \text{where } k \text{ is an arbitrary constant.}$$

Now consider $\alpha_2 = -2$. Similarly, we get

$$\mathbf{v}_{\alpha_2} = \begin{bmatrix} -\frac{1}{2}k \\ k \end{bmatrix}$$

The major use of eigenvalues and eigenvectors is that they allow us to “diagonalise” a square matrix. By definition, eigenvalues and their corresponding eigenvectors satisfy:

$$\mathbf{A}\mathbf{v}_{\alpha_i} = \mathbf{v}_{\alpha_i}\alpha_i, \quad \forall i = 1, \dots, n$$

which can be written in matrix form as

$$\mathbf{A}\mathbf{V} = \mathbf{V}\mathbf{D} \tag{4}$$

where $\mathbf{V}$ and $\mathbf{D}$ are $n \times n$ matrices defined by

$$\mathbf{V} = \begin{bmatrix} \mathbf{v}_{\alpha_1} & \mathbf{v}_{\alpha_2} & \cdots & \mathbf{v}_{\alpha_n} \end{bmatrix} \quad \text{and} \quad \mathbf{D} = \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_n \end{bmatrix}$$

Equation (4) implies that if $\det(\mathbf{V}) \neq 0$, we have

$$\mathbf{V}^{-1}\mathbf{A}\mathbf{V} = \mathbf{D} \tag{5}$$

Note that RHS of (5) is a diagonal matrix with the eigenvalues of $\mathbf{A}$.

Key takeaways: Applying (5) to a square matrix $\mathbf{A}$ is a procedure called “diagonalisation.”

Proposition 1.3.6. *Let $\mathbf{A}$ be an $n \times n$ square matrix with eigenvalues $\alpha_1, \alpha_2, \dots, \alpha_n$. Then*

1. $\alpha_1 + \alpha_2 + \cdots + \alpha_n = \text{trace}(\mathbf{A})$
2. $\alpha_1 \cdot \alpha_2 \cdots \alpha_n = \det(\mathbf{A})$

This result will be useful to know the signs of the eigenvalues of $\mathbf{A}$, without computing the eigenvalues.

Example 1.3.7. Suppose that $\mathbf{A}$ is a 2×2 matrix. By (2) of Proposition 1.3.6, if $\det(\mathbf{A}) < 0$, $\mathbf{A}$ has two eigenvalues of opposite signs.

Recall that we are seeking to characterise the solution to the 2×2 ODE system:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

where $\mathbf{A}$ is a 2×2 nondiagonal matrix.

Suppose we apply the change of variables:

$$\mathbf{y}(t) = \mathbf{V}^{-1}\mathbf{x}(t) \tag{6}$$

where $\mathbf{V}$ is the matrix of eigenvectors of $\mathbf{A}$, as defined previously, i.e.,

$$\mathbf{V} = \begin{bmatrix} \mathbf{v}_{\alpha_1} & \mathbf{v}_{\alpha_2} \end{bmatrix}$$

By (6), it follows that

$$\dot{\mathbf{y}}(t) = \mathbf{V}^{-1}\dot{\mathbf{x}}(t) = \mathbf{V}^{-1}\mathbf{A}\mathbf{x}(t) = \mathbf{V}^{-1}\mathbf{A}\mathbf{V}\mathbf{y}(t) = \mathbf{D}\mathbf{y}(t)$$

where I applied (5) and the definition

$$\mathbf{D} = \begin{bmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{bmatrix}$$

So we end up with a diagonal system:

$$\dot{\mathbf{y}}(t) = \mathbf{D}\mathbf{y}(t)$$

Using this result, we can show the following proposition:

Proposition 1.3.8. *Consider the 2×2 ODE system:*

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

where $\mathbf{A}$ is a 2×2 matrix with 2 distinct real eigenvalues α_1 and α_2 . Then the general solution to the ODE system takes the form

$$\mathbf{x}(t) = C_1\mathbf{v}_{\alpha_1} \exp(\alpha_1 t) + C_2\mathbf{v}_{\alpha_2} \exp(\alpha_2 t) \tag{7}$$

where C_1 and C_2 are constants of integration.

Proof. Looking for solutions to the ODE system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \text{where } \mathbf{A} \text{ is non-diagonal.}$$

Using the change of variables $\mathbf{y} = \mathbf{V}^{-1}\mathbf{x}$ we get

$$\dot{\mathbf{y}} = \mathbf{D}\mathbf{y}.$$

Since this is a diagonal system, its solution is:

$$\begin{cases} y_1(t) = c_1 \exp(\alpha_1 t), \\ y_2(t) = c_2 \exp(\alpha_2 t). \end{cases}$$

Now recall that $\mathbf{y} = \mathbf{V}^{-1}\mathbf{x} \Rightarrow \mathbf{x} = \mathbf{V}\mathbf{y}$. Then

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{v}_{\alpha_1} & \mathbf{v}_{\alpha_2} \end{bmatrix} \begin{bmatrix} c_1 \exp(\alpha_1 t) \\ c_2 \exp(\alpha_2 t) \end{bmatrix} \Rightarrow \mathbf{x}(t) = \mathbf{v}_{\alpha_1} c_1 \exp(\alpha_1 t) + \mathbf{v}_{\alpha_2} c_2 \exp(\alpha_2 t).$$

□

By inspection of (7), it is clear that the dynamics and stability of the system are determined by the signs of the eigenvalues α_1 and α_2 .

1. If $\alpha_1 < 0$ and $\alpha_2 < 0$, the steady state $\mathbf{x}^* = \mathbf{0}$ is stable.
2. If $\alpha_1 > 0$ and $\alpha_2 > 0$, the steady state $\mathbf{x}^* = \mathbf{0}$ is unstable.
3. If α_1 and α_2 have opposite signs, the steady state $\mathbf{x}^* = \mathbf{0}$ is saddle-path stable.

This is a generalisation of the analysis of diagonal systems.

Example 1.3.9 (Homework). Show that if $\mathbf{A}$ is diagonal, we have $\alpha_1 = a_{11}$ and $\alpha_2 = a_{22}$.

Proof. Assume $\mathbf{A}$ is a 2×2 diagonal matrix,

$$\mathbf{A} = \begin{pmatrix} a_{11} & 0 \\ 0 & a_{22} \end{pmatrix}.$$

By the definition of eigenvalues in the notes, a scalar α is an eigenvalue of $\mathbf{A}$ if and only if it solves the characteristic equation

$$\det(\mathbf{A} - \alpha \mathbf{I}) = 0.$$

Compute

$$\mathbf{A} - \alpha \mathbf{I} = \begin{pmatrix} a_{11} - \alpha & 0 \\ 0 & a_{22} - \alpha \end{pmatrix},$$

so

$$\det(\mathbf{A} - \alpha \mathbf{I}) = (a_{11} - \alpha)(a_{22} - \alpha).$$

Setting this equal to zero gives

$$(a_{11} - \alpha)(a_{22} - \alpha) = 0 \quad \Rightarrow \quad \alpha = a_{11} \text{ or } \alpha = a_{22}.$$

Hence the two eigenvalues are a_{11} and a_{22} . Labeling them, we may take $\alpha_1 = a_{11}$ and $\alpha_2 = a_{22}$ (the order is arbitrary). This is exactly what the homework asks to show. □

The previous analysis on stability generalises to nonhomogeneous systems of the form

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{b}$$

where $\mathbf{b}$ is a vector of constants.

Specifically, the dynamics and stability of the system are determined by the signs of the eigenvalues of $\mathbf{A}$, as discussed in the previous slide. But careful: The solution is no longer given by (7).

So far we mainly dealt with solving and analysing stability of 2×2 systems of linear ODEs of the form:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

Practically, we can follow a “cookbook” procedure:

- Compute the eigenvalues and corresponding eigenvectors of $\mathbf{A}$, use that

$$\mathbf{x}(t) = C_1 \mathbf{v}_{\alpha_1} \exp(\alpha_1 t) + C_2 \mathbf{v}_{\alpha_2} \exp(\alpha_2 t)$$

and pin down the constants C_1 and C_2 via initial or boundary conditions.

- If we are only interested in assessing stability, we only need to evaluate the signs of the eigenvalues of $\mathbf{A}$.

1.4 Systems of Nonlinear ODEs

Next we generalise the analysis of linear ODE systems to nonlinear systems. We centre attention on two-dimensional systems of nonlinear ODEs of the form:

$$\dot{x}_1(t) = F_1(x_1(t), x_2(t))$$

$$\dot{x}_2(t) = F_2(x_1(t), x_2(t))$$

where F_1 and F_2 are nonlinear, and have continuous first derivatives.

In matrix form, the system above can be written as

$$\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t))$$

How can we solve this system? One way to proceed is by performing a graphical analysis. With two-dimensional systems of this form, this is always possible: we just use phase diagrams. The same methods for constructing phase diagrams in linear models apply here. This would provide a qualitative approach to evaluating the solution to the nonlinear system.

Graphical solutions, though, have two main drawbacks:

- They cannot be applied to analyse larger systems (beyond two dimensions).
- They do not provide quantitative answers (e.g., What would be the level of output per capita in country XX in 10 years?)

To address these issues, we aim at obtaining analytical solutions. Things are trickier analytically, though, because nonlinear systems rarely admit closed form solutions. However, we can always study linear approximations to the nonlinear system!

Steps to approximate a solution to $\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t))$ analytically:

1. Compute the steady state $\mathbf{x}^*$
2. Linearise the system around $\mathbf{x}^*$
3. Solve the linearised system

Warning: This procedure only provides an approximation to the solution in a neighbourhood around the steady state. That's why this approach is called a "Local" analysis.

To apply step 2, we need to apply Taylor-series expansions.

Theorem 1.4.1 (Taylor's Theorem in $\mathbb{R}$). *Let $f : [a, b] \rightarrow \mathbb{R}$ be a C^{n+1} function, and let $x^* \in (a, b)$. Then we can write f as*

$$f(x) = f(x^*) + \sum_{k=1}^n \frac{f^{(k)}(x^*)}{k!} (x - x^*)^k + R_n(x; x^*)$$

where

$$\frac{R_n(x; x^*)}{(x - x^*)^n} \rightarrow 0 \quad \text{as } x \rightarrow x^*.$$

Remark. • $f^{(k)}$ denotes the k -th order derivative of f .

- f is said to be C^n if f has continuous derivatives up to order n (i.e., if $f^{(k)}$ exists and it is continuous for $k = 1, \dots, n$).
- R_n is a “residual” which gets “very small” in a neighbourhood of x^* .

Definition 1.4.2. The n -th order polynomial

$$P_n(x; x^*) \equiv f(x^*) + \sum_{k=1}^n \frac{f^{(k)}(x^*)}{k!} (x - x^*)^k$$

is called the n -th order Taylor-Series Expansion of f around x^* .

By Taylor’s Theorem, we can approximate $f(x)$ around x^* as

$$f(x) \approx P_n(x; x^*)$$

Special case: If we use a first-order Taylor expansion (i.e., if $n = 1$), the approximation simplifies to

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

This is called a linearisation of f around x^* .

Taylor’s theorem generalises to functions of more than one variable.

Example 1.4.3. Let $f(x_1, x_2)$ be a C^1 function. Then we can linearise f around (x_1^*, x_2^*) as follows:

$$f(x_1, x_2) \approx f(x_1^*, x_2^*) + \frac{\partial f(x_1^*, x_2^*)}{\partial x_1} (x_1 - x_1^*) + \frac{\partial f(x_1^*, x_2^*)}{\partial x_2} (x_2 - x_2^*)$$

Recall that we aim to solve the systems of nonlinear ODEs of the form:

$$\dot{x}_1(t) = F_1(x_1(t), x_2(t))$$

$$\dot{x}_2(t) = F_2(x_1(t), x_2(t))$$

where F_1 and F_2 are nonlinear and C^1 functions.

Applying Taylor’s theorem, we can study the system’s dynamics around the steady state (x_1^*, x_2^*) .

Start by linearising F_1 and F_2 around (x_1^*, x_2^*) :

$$\dot{x}_1(t) \approx F_1(x_1^*, x_2^*) + \frac{\partial F_1(x_1^*, x_2^*)}{\partial x_1(t)} (x_1(t) - x_1^*) + \frac{\partial F_1(x_1^*, x_2^*)}{\partial x_2(t)} (x_2(t) - x_2^*)$$

$$\dot{x}_2(t) \approx F_2(x_1^*, x_2^*) + \frac{\partial F_2(x_1^*, x_2^*)}{\partial x_1(t)} (x_1(t) - x_1^*) + \frac{\partial F_2(x_1^*, x_2^*)}{\partial x_2(t)} (x_2(t) - x_2^*)$$

Using that $F_1(x_1^*, x_2^*) = F_2(x_1^*, x_2^*) = 0$ (steady state), we can write the linear approximation in matrix form as

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} \approx \begin{bmatrix} \frac{\partial F_1(x_1^*, x_2^*)}{\partial x_1(t)} & \frac{\partial F_1(x_1^*, x_2^*)}{\partial x_2(t)} \\ \frac{\partial F_2(x_1^*, x_2^*)}{\partial x_1(t)} & \frac{\partial F_2(x_1^*, x_2^*)}{\partial x_2(t)} \end{bmatrix} \begin{bmatrix} x_1(t) - x_1^* \\ x_2(t) - x_2^* \end{bmatrix}$$

Or

$$\dot{\mathbf{x}}(t) \approx \mathbf{J}_F(\mathbf{x}^*) (\mathbf{x}(t) - \mathbf{x}^*)$$

where $\mathbf{J}_F(\mathbf{x}^*)$ is the Jacobian of F at $\mathbf{x}^*$.

This is a (nonhomogeneous) system of linear ODEs. Consider the system of nonlinear ODEs $\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t))$. We can analyse the dynamics of the system in a neighbourhood of the steady state $\mathbf{x}^*$ by following these steps:

1. Compute the steady state $\mathbf{x}^*$
2. Linearise the system around $\mathbf{x}^*$ using

$$\dot{\mathbf{x}}(t) \approx \mathbf{J}_F(\mathbf{x}^*)(\mathbf{x}(t) - \mathbf{x}^*)$$

3. Solve the linearised system and/or evaluate signs of eigenvalues of $\mathbf{J}_F(\mathbf{x}^*)$ to assess stability

Techniques for solving ODE systems apply to analysing higher-order equations.

Example 1.4.4. Consider the second-order ODE:

$$\ddot{x}(t) = f(t, \dot{x}(t), x(t))$$

where $\ddot{x}(t) \equiv d^2x(t)/dt^2$. Define the new variable $y = \dot{x}$. Then I am left with the two-equation ODE system:

$$\dot{y}(t) = f(t, y(t), x(t))$$

$$\dot{x}(t) = y(t)$$

... which we can analyse using the techniques we studied previously.

2 Dynamic Optimisation in Continuous Time

2.1 Finite-Horizon Problems

A typical continuous-time optimisation problem (a.k.a. “optimal control problem”) with finite horizon looks like this:

$$\max_{\mathbf{x}(t), \mathbf{y}(t)} \int_0^T f(t, \mathbf{x}(t), \mathbf{y}(t)) dt \quad (\star)$$

subject to

$$\dot{\mathbf{x}}(t) = g(t, \mathbf{x}(t), \mathbf{y}(t))$$

$$\mathbf{x}(t) \in \mathcal{X} \text{ and } \mathbf{y}(t) \in \mathcal{Y} \text{ for all } t, \text{ and } \mathbf{x}(0) = \mathbf{x}_0$$

where:

- $\mathbf{x}$ is a vector of state variables, $\mathbf{y}$ is a vector of control variables.
- f and g are continuously differentiable in their arguments.
- $T < \infty$.

How do we solve $(\star)$? Or how do we characterise its solution? Key challenges:

- We are choosing over infinitely dimensional objects, i.e., the functions $\mathbf{x} : [0, T] \rightarrow \mathcal{X}$ and $\mathbf{y} : [0, T] \rightarrow \mathcal{Y}$.

- The constraints include a differential equation, rather than a set of equalities/inequalities.

Next, we present a method that overcomes these challenges: Pontryagin's Maximum Principle.

To simplify the exposition, we focus on the case in which both the state and the control variables are one dimensional. That is, $x : [0, T] \rightarrow \mathcal{X} \subset \mathbb{R}$ and $y : [0, T] \rightarrow \mathcal{Y} \subset \mathbb{R}$.

Definition 2.1.1. We start by defining the Hamiltonian function:

$$H(t, x(t), y(t), \lambda(t)) \equiv f(t, x(t), y(t)) + \lambda(t) g(t, x(t), y(t))$$

where $\lambda(t)$ is a C^1 function called the costate variable.

The Hamiltonian is akin to a Lagrangian function, and it is useful to formulate the Maximum Principle. But what does it represent?

- Changing the value of the control $y(t)$ has two effects on the objective:
 1. $y(t)$ directly affects current payoffs.
 2. $y(t)$ indirectly affects future payoffs, through the change in $x(t)$.
- The Hamiltonian captures both effects by introducing a new variable $\lambda(t)$, which can be thought of as the shadow price of increasing $x(t)$.

Theorem 2.1.2 (Maximum Principle: A Simplified Version). *Suppose that the problem $(\star)$ has an interior continuous solution $(\hat{x}(t), \hat{y}(t)) = [\hat{x}(t), \hat{y}(t)]_{t=0}^T$. Then there exists a continuously differentiable function $\lambda(t)$ such that $(\hat{x}(t), \hat{y}(t))$ satisfy:*

$$H_y(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0 \quad \forall t \in [0, T] \quad (1)$$

$$-H_x(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = \dot{\lambda}(t) \quad \forall t \in [0, T] \quad (2)$$

$$g(t, \hat{x}(t), \hat{y}(t)) = \dot{x}(t) \quad \forall t \in [0, T] \quad (3)$$

and the transversality condition

$$\lambda(T) = 0. \quad (4)$$

Proof. A formal proof of the Maximum Principle is beyond the scope of this course. A heuristic proof:

Set up the Lagrangian

$$\mathcal{L}[x, y, \lambda] = \int_0^T \left[f(t, x(t), y(t)) + \lambda(t) (g(t, x(t), y(t)) - \dot{x}(t)) \right] dt.$$

Equivalently,

$$\mathcal{L}[x, y, \lambda] = \int_0^T \left[H(t, x(t), y(t), \lambda(t)) - \lambda(t) \dot{x}(t) \right] dt.$$

Now let $(\hat{x}, \hat{y}, \lambda)$ be an optimal triple and consider admissible variations

$$x_\varepsilon(t) = \hat{x}(t) + \varepsilon \eta(t), \quad y_\varepsilon(t) = \hat{y}(t) + \varepsilon \xi(t), \quad \lambda_\varepsilon(t) = \lambda(t) + \varepsilon \zeta(t),$$

where $\eta(0) = 0$, and where the endpoint value $\eta(T)$ must satisfy the relevant endpoint restriction. For notational simplicity, suppress hats in the computation below. The first variation is

$$0 = \frac{d}{d\varepsilon} \mathcal{L}[x_\varepsilon, y_\varepsilon, \lambda_\varepsilon] \Big|_{\varepsilon=0} = \int_0^T \left[H_x \eta + H_y \xi + (g - \dot{x}) \zeta - \lambda \dot{\eta} \right] dt.$$

Integrating the last term by parts gives

$$\int_0^T -\lambda(t) \dot{\eta}(t) dt = -\lambda(T) \eta(T) + \lambda(0) \eta(0) + \int_0^T \dot{\lambda}(t) \eta(t) dt.$$

Since $x(0) = x_0$ is fixed, we have $\eta(0) = 0$. Hence

$$0 = \int_0^T \left[(H_x + \dot{\lambda}) \eta + H_y \xi + (g - \dot{x}) \zeta \right] dt - \lambda(T) \eta(T).$$

Because the interior variations η, ξ, ζ are arbitrary on $(0, T)$, we obtain the usual first-order conditions:

$$H_y = 0, \quad \dot{\lambda} = -H_x, \quad \dot{x} = g.$$

It remains to analyze the boundary term

$$-\lambda(T) \eta(T).$$

□

Ignore the superscript $\hat{}$ and leave implicit the dependency on t . By (1) we can write $y = Y(t, x, \lambda)$. Using into (2)-(4) and combining with $x(0) = x_0$ we get:

$$\dot{x} = g(t, x, Y(t, x, \lambda)) \tag{5}$$

$$\dot{\lambda} = -H_x(t, x, Y(t, x, \lambda), \lambda) \tag{6}$$

$$x(0) = x_0, \quad \lambda(T) = 0. \tag{7}$$

(4)-(6) is a system of ODEs in x and λ ! This is typically a nonlinear system, which can be analysed using the methods studied so far.

Depending on the application, the problem might also feature **endpoint constraints**. These are constraints on $x(T)$, the terminal value of the state variable.

How should we modify the Maximum Principle theorem? We consider two possible cases:

$$1. \ x(T) = x_T$$

$$2. \ x(T) \geq x_T$$

In either case, our previous result holds, but we only modify the transversality condition $\lambda(T) = 0$ from equation (4).

Endpoint Constraint	Transversality Condition
None ($x(T)$ is free)	$\lambda(T) = 0$
$x(T) = x_T$	None
$x(T) \geq x_T$	$\lambda(T)(x(T) - x_T) = 0$

Proof. Case 1: $x(T)$ is free. In this case, $\eta(T)$ is arbitrary. Therefore the boundary term must vanish for every $\eta(T)$, which implies

$$\lambda(T) = 0.$$

Case 2: $x(T) = x_T$. Since the terminal state is fixed, every admissible variation must satisfy

$$\eta(T) = 0.$$

Hence the boundary term vanishes automatically:

$$-\lambda(T)\eta(T) = 0.$$

Therefore there is *no additional transversality condition*. The endpoint restriction $x(T) = x_T$ is already part of the problem itself.

Case 3: $x(T) \geq x_T$. There are two subcases.

First, if $\hat{x}(T) > x_T$, then the endpoint constraint is slack. For small perturbations, the terminal value is locally free, so $\eta(T)$ can be chosen with either sign. Thus, exactly as in Case 1,

$$\lambda(T) = 0.$$

Therefore,

$$\lambda(T)(\hat{x}(T) - x_T) = 0.$$

Second, if $\hat{x}(T) = x_T$, then the endpoint constraint binds. In that case,

$$\hat{x}(T) - x_T = 0,$$

and therefore

$$\lambda(T)(\hat{x}(T) - x_T) = 0$$

holds automatically, regardless of the value of $\lambda(T)$.

Combining the two subcases yields

$$\lambda(T)(\hat{x}(T) - x_T) = 0.$$

This proves the three endpoint/transversality cases. □

Important distinction:

- An **endpoint constraint** is imposed on endpoint values of the state variables, as part of the formulation of the problem.
- A **transversality condition** is a necessary condition for optimality imposed on the endpoint values of the states and/or costates.

Theorem 2.1.3 (Maximum Principle in Finite-Horizon). *Consider the finite-horizon problem $(\star)$, but suppose that the problem also features endpoint constraints on $x(T)$. Suppose that the problem has a continuous solution $(\hat{x}(t), \hat{y}(t))$. Then there exists a continuously differentiable function $\lambda(t)$ such that*

$$H_y(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0, \quad \forall t \in [0, T].$$

$$H_x(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = -\dot{\lambda}(t), \quad \forall t \in [0, T].$$

$$g(t, \hat{x}(t), \hat{y}(t)) = \dot{x}(t), \quad \forall t \in [0, T].$$

and the transversality condition

$$\lambda(T) = 0 \quad \text{if } x(T) \text{ is free.}$$

$$\lambda(T)(\hat{x}(T) - x_T) = 0 \quad \text{if } x(T) \geq x_T.$$

The Maximum Principle only provides necessary conditions for an optimum. Sufficient conditions for a maximum rely on concavity properties. We can show the following:

- If the Hamiltonian is concave in (x, y) , then the conditions of the Maximum Principle are also sufficient for a global maximum.
- If $H(t, x, y, \lambda)$ is strictly concave in (x, y) , then the solution which satisfies the Maximum Principle is unique.

In this course, sufficient conditions will typically be satisfied.

2.2 Infinite-Horizon Problems

A typical infinite-horizon optimal control problem looks like this:

$$\max_{\mathbf{x}(t), \mathbf{y}(t)} \int_0^\infty f(t, \mathbf{x}(t), \mathbf{y}(t)) dt \quad (8)$$

subject to

$$\begin{aligned} \dot{\mathbf{x}}(t) &= g(t, \mathbf{x}(t), \mathbf{y}(t)) \\ \mathbf{x}(t) &\in \mathcal{X} \quad \text{and} \quad \mathbf{y}(t) \in \mathcal{Y} \quad \text{for all } t, \quad \text{and} \quad \mathbf{x}(0) = \mathbf{x}_0 \end{aligned}$$

and some endpoint constraint on $\mathbf{x}(t)$ as $t \rightarrow \infty$.

Infinite-horizon problems present additional technicalities compared to finite-horizon problems. Such technical difficulties mainly arise because the objective function may not be finite.

In what follows, we circumvent most of those issues by focusing on a particular class of infinite-horizon problems: **Discounted Infinite-Horizon Problems**. All the problems we study in this course can be formulated as discounted infinite-horizon problems.

2.2.1 Exponential Discounting

Suppose that the decision maker solving the problem (8) discounts future payoffs exponentially at a constant rate $\rho > 0$. Then the objective function takes the form:

$$V = \int_0^\infty \exp(-\rho t) f(x(t), y(t)) dt \quad (9)$$

where

- $f(x(t), y(t))$ is the current-value of choosing $(x(t), y(t))$ in t
- $\exp(-\rho t)f(x(t), y(t))$ is the discounted-value of choosing $(x(t), y(t))$ in t , expressed in $t = 0$ terms

Note: Payoff function depends on t only through exponential discounting.

We can motivate (9) as follows: Suppose we are in discrete time $t = 0, 1, 2, \dots$. Exponential discounting at the rate ρ in this context means that the objective function can be written as

$$\begin{aligned} V &= f(x(0), y(0)) + \frac{1}{1+\rho} f(x(1), y(1)) + \left(\frac{1}{1+\rho}\right)^2 f(x(2), y(2)) + \dots \\ &= \sum_{t=0}^{\infty} \left(\frac{1}{1+\rho}\right)^t f(x(t), y(t)) \end{aligned} \quad (10)$$

It is easy to show that (9) is the continuous-time limit of (10). (see Acemoglu Section 5.5.)

2.2.2 Discounted Infinite-Horizon Problems

Our main interest in this course is to solve discounted infinite-horizon problems of this form:

$$\max_{x(t), y(t)} \int_0^{\infty} \exp(-\rho t) f(x(t), y(t)) dt \quad (\star\star)$$

subject to

$$\begin{aligned} \dot{x}(t) &= g(t, x(t), y(t)) \\ x(t) &\in \mathcal{X} \quad \text{and} \quad y(t) \in \mathcal{Y} \quad \text{for all } t, \quad x(0) = x_0 \end{aligned}$$

and an endpoint constraint

$$\lim_{t \rightarrow \infty} \exp(-\rho t) x(t) \geq 0.$$

Interpretation of the endpoint constraint: The discounted value of the state variable by the end of the planning horizon should be positive. In the realm of consumer's optimisation, this will map into a no-Ponzi condition.

The conditions of the Maximum Principle in finite-horizon extend to discounted infinite-horizon problems, such as $(\star\star)$. But we need to be careful with the transversality condition...

Let's start by writing the Hamiltonian associated with $(\star\star)$, following the insights from the finite-horizon analysis:

$$H(t, x, y, \lambda) = \exp(-\rho t) f(t, x, y) + \lambda g(t, x, y).$$

If we define $\mu(t) \equiv \exp(\rho t)\lambda(t)$, the above can be written as

$$H(t, x, y, \mu) = \exp(-\rho t) [f(t, x, y) + \mu g(t, x, y)] = \exp(-\rho t) \hat{H}(t, x, y, \mu).$$

The expression

$$\hat{H}(t, x, y, \mu) \equiv f(t, x, y) + \mu g(t, x, y)$$

is called the current-value Hamiltonian.

We can write the conditions of the Maximum Principle using either the original Hamiltonian H , or the current-value Hamiltonian $\hat{H}$. But we use the latter, which is the standard in the literature of economic growth. It also simplifies the algebra a bit, as we don't need to carry the discounting $\exp(-\rho t)$ throughout the calculations.

Theorem 2.2.1 (Maximum Principle in Discounted Infinite-Horizon). *Consider the discounted infinite-horizon problem $(\star\star)$. Suppose that the problem has a continuous solution $(\hat{x}(t), \hat{y}(t))$. Then there exists a continuously differentiable function $\mu(t)$ such that*

$$\hat{H}_y(t, \hat{x}(t), \hat{y}(t), \mu(t)) = 0 \quad \forall t \in [0, \infty)$$

$$\hat{H}_x(t, \hat{x}(t), \hat{y}(t), \mu(t)) = \rho\mu(t) - \dot{\mu}(t) \quad \forall t \in [0, \infty)$$

$$g(t, \hat{x}(t), \hat{y}(t)) = \dot{x}(t) \quad \forall t \in [0, \infty)$$

and the transversality condition

$$\lim_{t \rightarrow \infty} \exp(-\rho t) \mu(t) x(t) = 0.$$

Example 2.2.2. Homework: Check that the first two conditions coincide with those under finite-horizon if we used the original Hamiltonian H .

Proof. First, differentiate H with respect to y :

$$H_y(t, x, y, \lambda) = e^{-\rho t} \hat{H}_y(t, x, y, \mu),$$

since the factor $e^{-\rho t}$ does not depend on y . Therefore,

$$H_y = 0 \iff \hat{H}_y = 0.$$

Next, consider the costate equation in terms of the original Hamiltonian:

$$H_x = -\dot{\lambda}.$$

Since

$$H_x = e^{-\rho t} \hat{H}_x$$

and

$$\lambda(t) = e^{-\rho t} \mu(t),$$

we have

$$\dot{\lambda}(t) = \frac{d}{dt} (e^{-\rho t} \mu(t)) = e^{-\rho t} (\dot{\mu}(t) - \rho \mu(t)).$$

Substituting into $H_x = -\dot{\lambda}$ gives

$$e^{-\rho t} \hat{H}_x = -e^{-\rho t} (\dot{\mu} - \rho \mu).$$

Multiplying both sides by $e^{\rho t}$, we obtain

$$\hat{H}_x = \rho \mu - \dot{\mu}.$$

□

Note that we imposed the transversality condition:

$$\lim_{t \rightarrow \infty} \exp(-\rho t) \mu(t) x(t) = 0 \quad (\text{TVC})$$

FYI, there are weaker versions of this conditions which hold more generally. e.g., if we didn't have discounting, we would need a different transversality condition.

In the models we will consider, however, it will always be valid to use the condition (TVC). Such condition is also easier to check than other versions.

The transversality condition (TVC) can be thought of as the infinite-horizon version of its finite-horizon counterpart. Recall that if we imposed with $x(T) \geq 0$ in finite-horizon, the transversality condition is

$$\lambda(T)x(T) = 0.$$

Taking limits:

$$\lim_{t \rightarrow \infty} \lambda(t)x(t) = 0.$$

Using that $\lambda(t) = \exp(-\rho t)\mu(t)$ gives

$$\lim_{t \rightarrow \infty} \exp(-\rho t) \mu(t) x(t) = 0$$

which is our transversality condition (TVC).

As before, the Maximum Principle in infinite-horizon only provides necessary conditions for a maximum. Sufficiency conditions in infinite horizon rely not only the concavity, but also on the behaviour of the solution as $t \rightarrow \infty$. We will assume that such sufficiency conditions hold, unless stated otherwise.

The techniques discussed in this Lecture readily generalise to:

- Multidimensional problems (where $\mathbf{x}$ and $\mathbf{y}$ are vectors),
- Incorporating within-period constraints (e.g. $x(t) + 2y(t) \geq 3$).

We will present such generalisations if needed, as we go along.

3 Neoclassical Growth Model

In this section we cover the Neoclassical Growth Model. We consider a continuous time version based on the contributions of Ramsey-Cass-Koopmans (around 1965). Big question underlying the model: What is the role of capital accumulation as an engine of economic growth? Other works around the time had already addressed such question, in particular the Solow-Swan model.

The main contribution of Ramsey-Cass-Koopmans model is to endogenise the savings rate by introducing household optimisation. In doing so, it can potentially account for empirical patterns in the savings rate (e.g., that savings rates increase as countries get richer).

Overall, the quantitative performance of the model is not great. In particular, it cannot account for the large income differences across countries observed in the data (see Chari, Kehoe, McGrattan (1997)).

However, the methodological contribution of the model is HUGE:

Sets the stage of Dynamic General Equilibrium (DGE) analysis. DGE is at the core of macroeconomics in general, and of other areas of economics as well.

In the words of Acemoglu, the Neoclassical Growth Model is “arguably the most important model in macroeconomics.” (!)

The Neoclassical Growth Model features a representative household. You may have studied an alternative framework which also endogenises savings rates, but with heterogeneous agents: The Overlapping Generations Model (OLG). No model is necessarily superior to the other, as this depends on the application at hand.

But today’s framework has some methodological advantages. E.g., there is no dynamic inefficiency (overaccumulation of capital), which renders the analysis more tractable in some ways.

3.1 The Model

Infinite horizon economy in continuous time indexed by $t \in [0, \infty)$. Two types of infinitely lived agents:

- Representative Firm
- Representative Household

The economy is closed (i.e., does not interact with the rest of the world, then $S = I$). Single good, which can be consumed or invested.

The firm produces output by operating the production function

$$Y(t) = F(K(t), L(t))$$

where

- $Y(t)$ is the level of output produced at date t
- $K(t)$ is the level of capital dedicated to production at date t
- $L(t)$ is the level of labour dedicated to production at date t

Assumption 3.1.1 (Neoclassical Technology). 1. Constant returns to scale:

$$F(\lambda K, \lambda L) = \lambda F(K, L) \quad \text{for any } \lambda > 0$$

2. Positive marginal products:

$$F_K(K, L) > 0 \quad \text{and} \quad F_L(K, L) > 0$$

3. Diminishing marginal products:

$$F_{KK}(K, L) < 0 \quad \text{and} \quad F_{LL}(K, L) < 0$$

4. Inada conditions:

$$\lim_{i \rightarrow 0} F_i(K, L) = \infty \quad \text{and} \quad \lim_{i \rightarrow \infty} F_i(K, L) = 0, \quad \text{with } i \in \{K, L\}$$

The economy starts with $K(0) = K_0 > 0$, which is endowed to the household. Capital depreciates

exponentially at the rate $\delta \in (0, 1)$. Then total capital stock evolves over time according to:

$$\dot{K}(t) = I(t) - \delta K(t) \quad (1)$$

where $I(t) = Y(t) - C(t)$ denotes investment (= saving). Equation (1) is the “law of motion” for capital.

The household lives forever. Household population grows exponentially at the rate n . Let the initial population level be $N(0) = 1$. Population at time t is then given by

$$N(t) = \exp(nt).$$

The household derives utility from the consumption good, and supplies labour inelastically. The total endowment of labour in the economy is the total level of the population. Hence, inelastic labour supply implies

$$L(t) = N(t)$$

for all t .

Instantaneous utility function per household member is

$$u(c(t)) \quad (2)$$

where

$$c(t) = \frac{C(t)}{L(t)}$$

is consumption per capita at date t .

Assumption 3.1.2 (Neoclassical Preferences). The instantaneous utility function satisfies $u'(c) > 0$ and $u''(c) < 0$.

Remark. The definition of (2) also implies:

- (i) Time-separability: $u(c(t))$ does not depend on $c(s)$ for $s \neq t$.
- (ii) Stationarity: the function u is time-independent.

The instantaneous utility across all household members is $L(t)u(c(t))$.

The household discounts future payoffs (measured by u) exponentially, according to the subjective discount factor $\rho > 0$. This implies that preferences over $[c(t)]_{t=0}^{\infty}$ are represented by the utility function:

$$U = \int_0^{\infty} \exp(-\rho t) L(t) u(c(t)) dt = \int_0^{\infty} \exp(-(\rho - n)t) u(c(t)) dt$$

Important distinction:

- u represents preferences over consumption within a given period.
- U represents preferences over the entire path of consumption $[c(t)]_{t=0}^{\infty}$.

We assume that $\rho > n$, so that there is discounting.

3.2 Optimal Growth

We aim at characterising the Pareto Optimal allocations in the economy. To this end, we set up the problem of a benevolent social planner, in which:

- Objective function $\rightarrow$ Household's utility U
- Constraints $\rightarrow$ Technology to transform capital into consumption

The planner does not use markets to allocate resources!

It will be convenient to formulate the planning problem in terms of *per capita* variables. Let

$$c(t) \equiv \frac{C(t)}{L(t)} \rightarrow \text{consumption per capita}$$

$$y(t) \equiv \frac{Y(t)}{L(t)} \rightarrow \text{output per capita}$$

$$k(t) \equiv \frac{K(t)}{L(t)} \rightarrow \text{capital per capita}$$

Note that the properties of the production function imply

$$y(t) \equiv \frac{F(K(t), L(t))}{L(t)} = F\left(\frac{K(t)}{L(t)}, 1\right) = f(k(t))$$

From now on, we refer to $f(k)$ as the “per capita production function.”

We also need to write the law of motion for capital in per capita terms. Recall:

$$\dot{K}(t) = Y(t) - C(t) - \delta K(t)$$

Then

$$\frac{\dot{K}(t)}{L(t)} = \frac{Y(t)}{L(t)} - \frac{C(t)}{L(t)} - \delta \frac{K(t)}{L(t)}$$

Lemma 3.2.1. *It is straightforward to show that*

$$\frac{\dot{K}(t)}{L(t)} = \dot{k}(t) + k(t)n.$$

Proof. By definition,

$$\dot{k}(t) := \frac{d(K(t)/L(t))}{dt} = \frac{\dot{K}(t)L(t) - K(t)\dot{L}(t)}{(L(t))^2} = \frac{\dot{K}(t)}{L(t)} - k(t)n \quad \Rightarrow \quad \frac{\dot{K}(t)}{L(t)} = \dot{k}(t) + k(t)n.$$

□

Hence, the law of motion of capital in per capita terms is

$$\dot{k}(t) = f(k(t)) - (\delta + n)k(t) - c(t)$$

The optimal growth problem can then be written as

$$\max_{[c(t), k(t)]_{t=0}^{\infty}} \int_0^{\infty} \exp(-(\rho - n)t) u(c(t)) dt$$

subject to

$$\dot{k}(t) = f(k(t)) - (\delta + n)k(t) - c(t)$$

$$k(0) = k_0 > 0$$

$$\lim_{t \rightarrow \infty} \exp(-(\rho - n)t) k(t) \geq 0$$

We also need $c(t) \geq 0$ and $k(t) \geq 0$ for all t , but these will always be satisfied by Inada conditions.

Current value Hamiltonian is

$$H(c, k, \mu) = u(c) + \mu [f(k) - (\delta + n)k - c]$$

Necessary conditions for optimality include

$$H_c = 0 \quad \Leftrightarrow \quad u'(c) - \mu = 0 \quad (3)$$

$$H_k = -\dot{\mu} + (\rho - n)\mu \quad \Leftrightarrow \quad \mu [f'(k) - \delta - n] = -\dot{\mu} + (\rho - n)\mu \quad (4)$$

$$\dot{k} = f(k) - (\delta + n)k - c \quad (5)$$

and the transversality condition

$$\lim_{t \rightarrow \infty} \exp(-(\rho - n)t) \mu(t) k(t) = 0 \quad (6)$$

We can show that sufficient conditions for a maximum hold. We want to eliminate μ , and write the system in terms of (c, k) .

Differentiating (3) with respect to t we get

$$u''(c) \dot{c} = \dot{\mu} \quad (7)$$

Using (3) and (7) into (4) gives

$$u'(c) [f'(k) - \delta - n] = -u''(c) \dot{c} + (\rho - n)u'(c)$$

A bit of algebra leads to **the consumption Euler Equation**:

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho}{\sigma(c)}$$

where

$$\frac{1}{\sigma(c)} \equiv -\frac{u'(c)}{u''(c)c}$$

is the intertemporal elasticity of substitution (IES). Intuition: it measures the willingness of the consumer to substitute consumption overtime as the intertemporal price of consumption changes. A two period example see Figure 4.

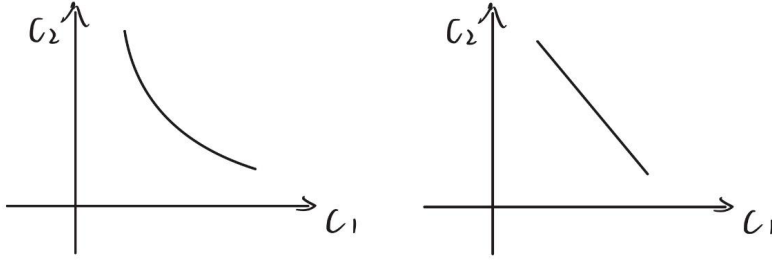


Figure 4: Left: high curvature, low IES. Right: low curvature, high IES.

From now on, we make the following assumption:

Assumption 3.2.2 (Constant Relative Risk Aversion). The instantaneous utility function u takes the form

$$u(c) = \begin{cases} \frac{c^{1-\sigma}}{1-\sigma}, & \text{if } \sigma \neq 1, \text{ and } \sigma > 0, \\ \log(c), & \text{if } \sigma = 1. \end{cases}$$

In this case:

$$\sigma(c) = \sigma \quad \text{for all } c$$

Under CRRA utility, the Euler Equation becomes:

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho}{\sigma}$$

Intuition:

- Consumption growth is proportional to the difference between the net return of investing in capital, and the discount factor.
- When such difference is positive, the net return on capital compensates consumer's impatience.
- In that case, it pays off to invest in physical capital, which decreases c today, but raises it in the future: i.e., consumption growth is positive.
- The IES $\frac{1}{\sigma}$ affects the sensitivity of consumption growth to changes in $(f'(k) - \delta - \rho)$.

The ODE system in terms of (c, k) boils down to

$$\frac{\dot{c}}{c} = \frac{f'(k) - \delta - \rho}{\sigma} \tag{8}$$

$$\dot{k} = f(k) - (\delta + n)k - c \tag{9}$$

along with the transversality condition

$$\lim_{t \rightarrow \infty} \exp(-(\rho - n)t) \mu(t) k(t) = 0 \tag{10}$$

and $k(0) = k_0$.

Let (c^*, k^*) be the steady state such that $\dot{c} = \dot{k} = 0$. Using (8)-(9), the steady state satisfies:

$$f'(k^*) = \delta + \rho \quad (11)$$

$$c^* = f(k^*) - (\delta + n)k^* \quad (12)$$

Note:

- Steady state values do not depend on the shape of the instantaneous utility function u .
- However, discounting plays an important role...

By (11), note that

$$f'(k^*) = \delta + \rho > \delta + n = f'(k^{GR}) \iff k^* < k^{GR}$$

where k^{GR} is the “Golden Rule” level of capital, i.e., the level of k that maximises steady state consumption. Formally,

$$k^{GR} = \arg \max_k \{f(k) - (n + \delta)k\} \Rightarrow f'(k^{GR}) = \delta + n.$$

k^* is referred to as the “Modified Golden Rule” level of k . Intuition: Due to discounting (i.e., $\rho - n > 0$), earlier consumption is preferred to later consumption. This insight is outside of the scope of the Solow model, which does not endogenise the savings rate.

3.2.1 Phase Diagram

The locus $\dot{c} = 0$ is found using (8). It is a vertical line at $k = k^*$, which is $f'(k^*) = \delta + \rho$. It also follows that

$$\dot{c} > 0 \iff f'(k) > \delta + \rho \iff k < k^*.$$

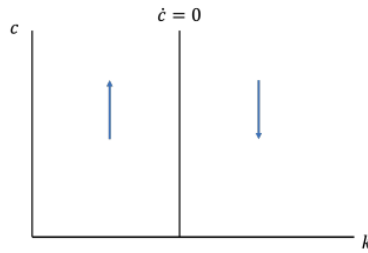


Figure 5: The locus $\dot{c} = 0$ is found using (8). It is a vertical line at $k = k^*$, which is $f'(k^*) = \delta + \rho$.

The locus $\dot{k} = 0$ is found using (9). It is a curve given by

$$c(k) = f(k) - (\delta + n)k.$$

It also follows that

$$\dot{k} > 0 \iff c(k) < f(k) - (\delta + n)k.$$

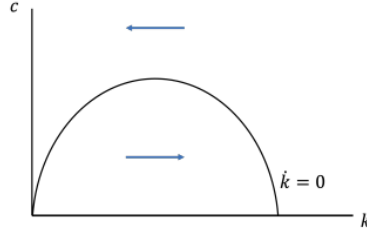


Figure 6: The locus $\dot{k} = 0$ is found using (9).

Qualitatively, the system displays saddle-path stability, so there is a unique path which is stable.

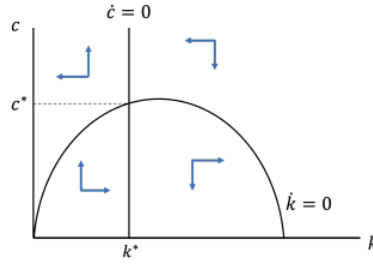


Figure 7: Qualitatively, the system displays saddle-path stability.

Given $k(0) = k_0$, the path of $(c(t), k(t))$ along the stable arm is also the unique path which solves the optimal growth problem.

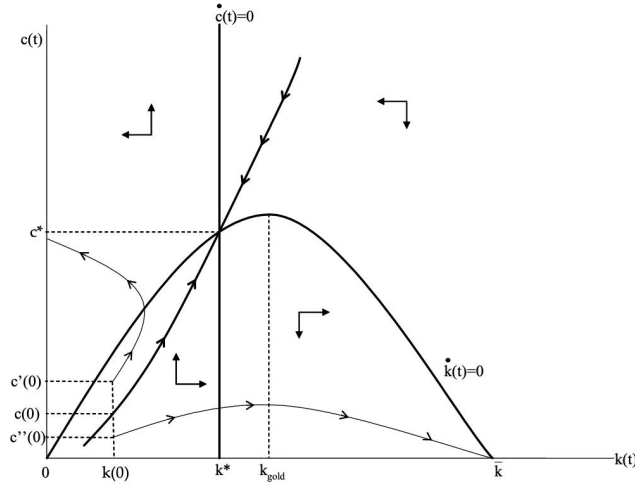


Figure 8: Given $k(0) = k_0$, the path of $(c(t), k(t))$ along the stable arm is also the unique path which solves the optimal growth problem.

Let $c(0)$ be the level of consumption on the stable arm, given $k(0)$.

- Suppose that initial consumption is $c'(0) < c(0)$.
 - Then $c(t) \rightarrow 0$ and $k(t) \rightarrow \bar{k} > k^{GR}$ (see phase diagram).

- This trajectory violates the transversality condition (intuitively: it cannot be optimal to overaccumulate capital while starving at the same time).
- Suppose that initial consumption is $c''(0) > c(0)$.
 - Then $c(t) \rightarrow \bar{c} > 0$ and $k(t) \rightarrow 0$ (see phase diagram).
 - This trajectory violates the feasibility!

Conclusion: Given $k(0) = k_0$, the path along the stable arm is the unique path which solves the optimal growth problem.

The savings rate is defined by

$$s \equiv \frac{f(k) - c}{f(k)} = 1 - \frac{c}{f(k)}.$$

This model admits rich patterns on how s evolves along the transition to the steady state. In particular, we can isolate parametric conditions such that the savings rate increases as per capita income rises along the transition. This is an empirical fact which cannot be rationalised in a model with exogenous savings rates, like the Solow model.

3.2.2 Local Analysis

Ignoring the transversality and initial conditions for a moment, the ODE system in (8)-(9) was

$$\begin{aligned}\dot{c}(t) &= \frac{f'(k(t)) - \delta - \rho}{\sigma} c(t) \equiv g_1(c(t), k(t)), \\ \dot{k}(t) &= f(k(t)) - (\delta + n)k(t) - c(t) \equiv g_2(c(t), k(t)).\end{aligned}$$

Linearising around (c^*, k^*) we get

$$\begin{bmatrix} \dot{c}(t) \\ \dot{k}(t) \end{bmatrix} \approx \underbrace{\begin{bmatrix} \frac{\partial g_1(c^*, k^*)}{\partial c^*} & \frac{\partial g_1(c^*, k^*)}{\partial k^*} \\ \frac{\partial g_2(c^*, k^*)}{\partial c^*} & \frac{\partial g_2(c^*, k^*)}{\partial k^*} \end{bmatrix}}_{\equiv \mathbf{J}_F(c^*, k^*)} \begin{bmatrix} c(t) - c^* \\ k(t) - k^* \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & \frac{f''(k^*)}{\sigma} c^* \\ -1 & \rho - n \end{bmatrix}}_{\equiv \mathbf{J}_F(c^*, k^*)} \begin{bmatrix} c(t) - c^* \\ k(t) - k^* \end{bmatrix},$$

where $\mathbf{J}_F(c^*, k^*)$ is the Jacobian of the system evaluated at (c^*, k^*) .

Using that $f'' < 0$, we get

$$\det(\mathbf{J}_F(c^*, k^*)) = \frac{f''(k^*)}{\sigma} c^* < 0$$

Hence, there are two real eigenvalues, (α_1, α_2) , with

$$\alpha_1 < 0 < \alpha_2$$

This confirms that the steady state is locally saddle-path stable.

We can go one step further and get a closed form solution to the linear approximation. Using the solution methods to linear ODEs discussed in previous Lectures, the general solution for k and c takes the form

$$\begin{aligned}c(t) - c^* &= \theta_1 v_{\alpha_1}^1 \exp(\alpha_1 t) + \theta_2 v_{\alpha_2}^1 \exp(\alpha_2 t) \\ k(t) - k^* &= \theta_1 v_{\alpha_1}^2 \exp(\alpha_1 t) + \theta_2 v_{\alpha_2}^2 \exp(\alpha_2 t)\end{aligned}$$

where $\mathbf{v}_{\alpha_i} = \begin{bmatrix} v_{\alpha_i}^1 & v_{\alpha_i}^2 \end{bmatrix}'$ with $i \in \{1, 2\}$ is the eigenvector associated with the i -th eigenvalue, and θ_1 and θ_2 are arbitrary constants.

Consider the general solution for $k(t)$. We can write this as

$$k(t) = k^* + \eta_1 \exp(\alpha_1 t) + \eta_2 \exp(\alpha_2 t) \quad (13)$$

where $\eta_i \equiv \theta_i v_{\alpha_i}^2$ for $i \in \{1, 2\}$.

To pin down η_1 and η_2 , note that

1. The initial condition $k(0) = k_0$ implies $k(0) = k^* + \eta_1 + \eta_2$.
2. We've also shown that the unique solution is on the stable arm. Hence, if $\alpha_1 < 0$, it must be that $\eta_2 = 0$.

Combining the above conditions leads to

$$k(t) = k^* + (k_0 - k^*) \exp(\alpha_1 t) \quad (14)$$

which is the optimal path for capital in the linearised model.

To recover the linearised solution for $c(t)$ we can proceed as follows:

1. The second equation of the linearised system is

$$\dot{k}(t) = -(c(t) - c^*) + (\rho - n)(k(t) - k^*) \quad (15)$$

2. Using (14) into (15) gives the equation for the stable arm in the linearised model

$$c(t) = c^* + (\rho - n - \alpha_1)(k(t) - k^*) \quad (16)$$

Using (14) into (16) gives

$$c(t) = c^* + (\rho - n - \alpha_1)(k_0 - k^*) \exp(\alpha_1 t) \quad (16.5)$$

which is the optimal path for consumption in the linearised model.

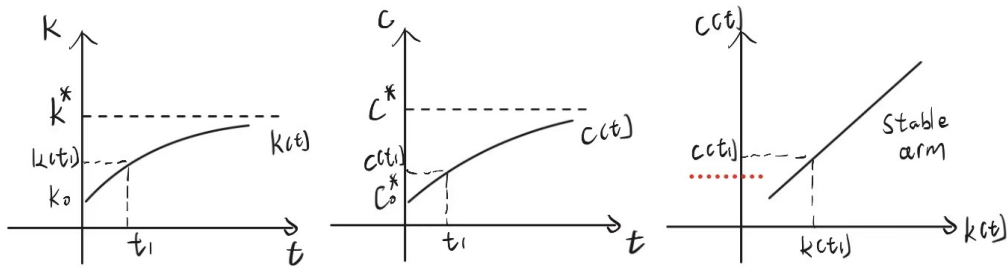


Figure 9: Left, equation (14); middle, equation (16.5); right, equation (16).

3.3 Competitive Equilibrium

Now we turn to analysing how resources are allocated using markets. Consumers and firms make decisions taking prices as given. We are mainly interested in how the resulting market allocation compares with the allocations of the optimal growth problem.

Key assumptions:

- Households own the factors of production (K, L) .
- All markets are competitive (households and firms are price takers).

Recall that the model features a representative firm living forever which produces output according to $F(K, L)$. The goal of the firm is maximise profits. The firm solves the following problem at each date t :

$$\max_{K^d(t), L^d(t)} F(K^d(t), L^d(t)) - R(t)K^d(t) - w(t)L^d(t)$$

where

- $(K^d(t), L^d(t))$ are capital and labour demand at t .
- $R(t)$ is the rental rate of capital.
- $w(t)$ is the wage.

First order conditions give

$$R(t) = F_K(K^d(t), L^d(t)), \quad w(t) = F_L(K^d(t), L^d(t)).$$

Or in per capita terms

$$R(t) = f'(k^d(t)), \quad w(t) = f(k^d(t)) - f'(k^d(t))k^d(t) \quad (17)$$

where we used that $F(K, L) = Lf(k) = Lf(\frac{K}{L})$ by CRS. Indeed,

$$F_K(K, L) = Lf'(\frac{K}{L})\frac{1}{L} = f'(\frac{K}{L}) \Rightarrow R(t) = f'(k^d(t)).$$

Similarly for the wage one.

The household consumes, works, and saves, to maximise utility U . Two assets are available for saving:

a riskless bond $B(t)$, and physical capital $K(t)$

- Riskless bond is in zero net supply (in balance) and yields the net interest rate $r(t)$.
- Physical capital is in positive net supply and is rented out to the firms each period at the rate $R(t)$.

The household faces the following constraints:

$$\dot{B}(t) = w(t)L(t) + r(t)B(t) + R(t)K(t) - C(t) - I(t)$$

$$\dot{K}(t) = I(t) - \delta K(t)$$

The first one is the household's budget constraint. The second one is the law of motion for capital. Combining these equations gives

$$\dot{B}(t) + \dot{K}(t) = w(t)L(t) + r(t)B(t) + (R(t) - \delta)K(t) - C(t) \quad (18)$$

If markets for bonds and capital work efficiently, the following non-arbitrage condition holds:

$$r(t) = R(t) - \delta$$

LHS: rate of return on bonds; RHS: rate of return on capital.

Reason: If $r(t) \neq R(t) - \delta$, the household could make a risk-free profit without costs by rebalancing its portfolio. This is not consistent with equilibrium.

Given the non-arbitrage condition $r(t) = R(t) - \delta$, we can write (18) as

$$\dot{B}(t) + \dot{K}(t) = w(t)L(t) + r(t)(B(t) + K(t)) - C(t)$$

or

$$\dot{A}(t) = w(t)L(t) + r(t)A(t) - C(t) \quad (19)$$

where $A(t) \equiv B(t) + K(t)$ denotes total assets.

Note: Since bonds are in zero net supply ($B(t) = 0$), in equilibrium we have

$$A(t) = K(t).$$

From now on, we refer to equation (19) as the household's budget constraint.

It will be convenient to write the household's budget constraint in per capita terms. Define $a(t) \equiv \frac{A(t)}{L(t)}$ as the per capita level of asset holdings. Then (19) implies

$$\dot{a}(t) = w(t) + (r(t) - n)a(t) - c(t) \quad (20)$$

where we used that

$$\frac{\dot{A}(t)}{L(t)} = \dot{a}(t) + a(t)n.$$

The household also faces an endpoint constraint, given by

$$\lim_{t \rightarrow \infty} \left[\exp \left(- \int_0^t (r(s) - n) ds \right) a(t) \right] \geq 0 \quad (21)$$

This is the so-called “No-Ponzi Game Condition.” In words, it requires that the present value of assets be positive in the long run. This prevents the household from continuously borrowing to pay previous debts. Otherwise, financial markets would incur losses in the long run.

Using (20) and (21), we can formulate the household's problem as

$$\max_{[c(t), a(t)]_{t=0}^{\infty}} \int_0^{\infty} \exp \left(- (\rho - n)t \right) u(c(t)) dt$$

subject to

$$\begin{aligned} \dot{a}(t) &= w(t) + (r(t) - n)a(t) - c(t), \\ a(0) &= a_0, \\ \lim_{t \rightarrow \infty} \left[\exp \left(- \int_0^t (r(s) - n) ds \right) a(t) \right] &\geq 0. \end{aligned}$$

Definition 3.3.1 (Competitive Equilibrium). A competitive equilibrium consists of paths $[c(t), k(t), a(t), w(t), r(t)]_{t=0}^{\infty}$ such that

1. The household solves its problem, taking $[w(t), r(t)]_{t=0}^{\infty}$ as given.

2. The firm maximises profits, taking $[w(t), r(t)]_{t=0}^{\infty}$ as given.
3. All markets clear.

Proposition 3.3.2 (First and Second Welfare Theorems). *The competitive equilibrium path $[c(t), k(t)]_{t=0}^{\infty}$ is identical to the corresponding path which solves the optimal growth problem.*

Proposition 1 has a number of important implications:

- It establishes that the neoclassical growth model satisfies the First Welfare Theorem (i.e., the CE is PO).
- It also establishes an even more powerful result: the neoclassical growth model satisfies the Second Welfare Theorem (i.e., the PO is a CE).
- It establishes unicity of the CE as a corollary. (Why?)

Part II

Dynamic General Equilibrium Models with Uncertainty

4 Dynamic Programming

4.1 Introduction

Example 4.1.1. How would you solve this problem (stochastic growth model)?

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

subject to

$$c_t + k_{t+1} \leq Ak_t^\alpha \theta_t, \text{ for all } t, k_0 \text{ given, } \ln(\theta_t) \sim_{iid} N(0, \sigma^2),$$

where c_t is consumption, k_t is capital, θ_t is a productivity shock.

We are looking for a solution of the form:

$$\{c_t^*(\boldsymbol{\theta}^t)\}_{t=0}^{\infty}, \quad \{k_{t+1}^*(\boldsymbol{\theta}^t)\}_{t=0}^{\infty},$$

where $\boldsymbol{\theta}^t = \{\theta_0, \theta_1, \dots, \theta_t\}$ is the history of shocks up to t .

Beyond some special cases, such solution is not analytical and should be approximated... Key challenge is that the original problem is LARGE!

- We are maximising over an infinite sequence of controls.
- We are subject to an infinite sequence of constraints.
- Each period's optimal decisions depend on the history of shocks and past decisions until that date (history dependence).

How can these issues be handled? We will focus on Dynamic Programming.

Basic idea: Break up the original problem into a series of small and more tractable problems. The smaller problems will be casted in a recursive way. Under certain regularity conditions, the solution to such recursive problems can be used to recover the solution to the original problem.

This is an application of **Bellman's Optimality Principle**: An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision. (Bellman, 1957)

Dynamic Programming is certainly not the only tool to analyse dynamic optimisation models. In general, methods to approximate solution to such problems (either in discrete or in continuous time) can be divided into two big groups:

Definition 4.1.2 (Perturbation methods). Find a particular case that has a known solution (e.g., a steady state). Approximate solution in a *neighbourhood* of such particular case.

This is what we did in continuous-time when linearising around the steady state. Similar methods apply in discrete time.

Definition 4.1.3 (Projection methods). Find a parametric function that is “close” to the exact solution *over the entire domain*.

This is what we will do next using Dynamic Programming, by exploiting the recursive structure of problem.

4.2 Basic concepts

Definition 4.2.1 (State Variable). Set of variables summarising “the state of the economy” at each point in time. i.e., agents facing the same state variables make the same decisions independently of the time period.

Definition 4.2.2 (Control Variable). Set of choice variables at each point in time.

Definition 4.2.3 (Value Function). Optimal value of the original problem, given the states.

Definition 4.2.4 (Policy Function). Optimal values for the control variables, given the states.

4.3 Finite Horizon Problems

Consider a decision-maker solving the deterministic problem:

$$\max_{\{\mathbf{u}_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \beta^t r(\mathbf{x}_t, \mathbf{u}_t) + W(\mathbf{x}_T) \quad \text{s.t.} \quad \mathbf{x}_{t+1} = g(\mathbf{x}_t, \mathbf{u}_t), \quad \forall t, \quad \mathbf{x}_0 \text{ given.}$$

Where $\beta \in (0, 1)$ is the discount factor, $\mathbf{x} \in \mathcal{X}$ is the state variable, $\mathbf{u} \in \mathcal{U}$ is the control variable, $r(\mathbf{x}_t, \mathbf{u}_t)$ is a concave return function, $g(\mathbf{x}_t, \mathbf{u}_t)$ describes the law of motion for the state, and $W(\mathbf{x}_T)$ is the terminal condition at $\mathbf{x}_T$.

Assume a convex constraint set, to ensure a maximum exists.

The Value Function $V_t(\mathbf{x})$ is the greatest feasible payoff from time t forward, if the state at time t is $\mathbf{x}_t$:

$$V_t(\mathbf{x}_t) = \max_{\{\mathbf{u}_s\}_{s=t}^{T-1}} \sum_{s=t}^{T-1} \beta^s r(\mathbf{x}_s, \mathbf{u}_s) + W(\mathbf{x}_T) \quad \text{s.t.} \quad \mathbf{x}_t \text{ given, } \mathbf{x}_{s+1} = g(\mathbf{x}_s, \mathbf{u}_s), \quad \forall s,$$

Given the structure of the problem, the value function satisfies the Bellman Equation:

$$V_t(\mathbf{x}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V_{t+1}(g(\mathbf{x}, \mathbf{u}))\}$$

with terminal condition $V_T(\mathbf{x}) = W(\mathbf{x})$.

Given $V_t(\mathbf{x})$, we define the Policy Function $\mathbf{U}_t(\mathbf{x})$ by:

$$\mathbf{U}_t(\mathbf{x}) = \arg \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V_{t+1}(g(\mathbf{x}, \mathbf{u}))\}.$$

The Bellman equation and the policy function are the building blocks for constructing the optimal solution, as we'll see next.

Original Problem:

$$\max_{\{\mathbf{u}_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \beta^t r(\mathbf{x}_t, \mathbf{u}_t) + W(\mathbf{x}_T) \quad \text{s.t.} \quad \mathbf{x}_{t+1} = g(\mathbf{x}_t, \mathbf{u}_t) \quad \forall t, \quad \mathbf{x}_0 \text{ given.}$$

... find a sequence of T controls.

Recursive Formulation:

$$V_t(\mathbf{x}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V_{t+1}(g(\mathbf{x}, \mathbf{u}))\} \quad \forall t.$$

... solve T problems with one control each.

Solution method:

1. Compute $\{V_t(\mathbf{x})\}_{t=0}^T$ and $\{\mathbf{U}_t(\mathbf{x})\}_{t=0}^{T-1}$ by backward induction, using the Bellman equation.

(1) Compute $V_T(\mathbf{x}) = W(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{X}$.

(2) For all $t = T-1, T-2, \dots, 0$, compute:

$$V_t(\mathbf{x}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V_{t+1}(g(\mathbf{x}, \mathbf{u}))\}$$

and

$$\mathbf{U}_t(\mathbf{x}) = \arg \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V_{t+1}(g(\mathbf{x}, \mathbf{u}))\}$$

for all $\mathbf{x} \in \mathcal{X}$. This gives the value and policy functions $\{V_t(\mathbf{x})\}_{t=0}^T$ and $\{\mathbf{U}_t(\mathbf{x})\}_{t=0}^{T-1}$.

2. Given $\mathbf{x}_0$, compute $\{\mathbf{u}_t^*\}_{t=0}^{T-1}$ and $\{\mathbf{x}_t^*\}_{t=0}^T$ by iterating forward on $\{\mathbf{U}_t(\mathbf{x})\}_{t=0}^{T-1}$.

We iterate forward, using $\mathbf{x}_0$ (given):

$$\mathbf{u}_0^* = \mathbf{U}_0(\mathbf{x}_0), \quad \mathbf{x}_1^* = g(\mathbf{x}_0, \mathbf{u}_0^*),$$

$$\mathbf{u}_1^* = \mathbf{U}_1(\mathbf{x}_1^*), \quad \mathbf{x}_2^* = g(\mathbf{x}_1^*, \mathbf{u}_1^*),$$

...

$$\mathbf{u}_t^* = \mathbf{U}_t(\mathbf{x}_t^*), \quad \mathbf{x}_{t+1}^* = g(\mathbf{x}_t^*, \mathbf{u}_t^*),$$

...

$$\mathbf{u}_{T-1}^* = \mathbf{U}_{T-1}(\mathbf{x}_{T-1}^*), \quad \mathbf{x}_T^* = g(\mathbf{x}_{T-1}^*, \mathbf{u}_{T-1}^*).$$

4.4 Infinite Horizon Problems

Now consider the problem:

$$\max_{\{\mathbf{u}_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t r(\mathbf{x}_t, \mathbf{u}_t) \quad \text{s.t.} \quad \mathbf{x}_{t+1} = g(\mathbf{x}_t, \mathbf{u}_t), \quad \mathbf{x}_0 \text{ given,}$$

where in addition to previous properties on parameters and functions, we assume $r(\cdot)$ is bounded.

How can we compute the solution $\{\mathbf{u}_t^*, \mathbf{x}_{t+1}^*\}_{t=0}^{\infty}$ by applying Dynamic Programming?

Let's try to extend the concepts of the finite horizon problem...

Bellman equation:

$$V(\mathbf{x}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V(g(\mathbf{x}, \mathbf{u}))\} \quad (*)$$

Equation (*) defines a functional equation in V , which is the unknown function.

Can we apply the backward induction algorithm to the infinite horizon framework? Below we look at solution methods that rely on the following theorem.

Theorem 4.4.1. *Under certain technical conditions:*

1. *The Bellman equation has a unique strictly concave solution V^* .*
2. *Let V^0 be any bounded and continuous VF, and let V^j satisfy*

$$V^j(\mathbf{x}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V^{j-1}(g(\mathbf{x}, \mathbf{u}))\} \quad \text{for } j \geq 1.$$

Then $V^ = \lim_{j \rightarrow \infty} V^j$.*

3. *There is a unique and time-invariant policy function $\mathbf{U}^*$, s.t. the solution to the original problem can be obtained by iterating on $\mathbf{u}_t^* = \mathbf{U}^*(\mathbf{x}_t)$.*
4. *Off the corners, the limiting value function V^* is differentiable with:*

$$V^{*'}(\mathbf{x}) = r_1(\mathbf{x}, \mathbf{U}(\mathbf{x})) + \beta V^{*'}(g(\mathbf{x}, \mathbf{U}(\mathbf{x}))) g_1(\mathbf{x}, \mathbf{U}(\mathbf{x})).$$

Proof. See Stokey, Lucas, Prescott (1989). □

We concentrate on two solution methods:

- Guess and Verify. It is analytical; can be solved with pencil and paper. Only works under very special cases.

1. Guess that the true value function has a particular parametric form $V^G(\mathbf{x}; \mathbf{A})$, where $\mathbf{A}$ is a vector of unknown parameters.

2. Compute:

$$\hat{V}(\mathbf{x}; \mathbf{A}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta V^G(g(\mathbf{x}, \mathbf{u}); \mathbf{A})\}.$$

3. If there exists parameters $\mathbf{A}^*$ s.t. $\hat{V}(\mathbf{x}; \mathbf{A}^*) = V^G(\mathbf{x}; \mathbf{A}^*)$, then $V^G(\mathbf{x}; \mathbf{A}^*)$ is the solution.

Relies on the uniqueness of the solution to the Bellman equation. Only works for very special cases.

- Value Function Iteration. It is computational; requires numerical tools. It is essentially a generalisation of Method #1.

1. Start from any bounded and continuous guess V^0 .
2. Iterate on

$$V^j(\mathbf{x}) = \max_{\mathbf{u}} \left\{ r(\mathbf{x}, \mathbf{u}) + \beta V^{j-1}(g(\mathbf{x}, \mathbf{u})) \right\}$$

until convergence (e.g., until $\|V^j - V^{j-1}\|_\infty < \varepsilon$).

This method always works thanks to Property #2:

$$V^* = \lim_{j \rightarrow \infty} V^j.$$

A disadvantage is that this method can be slow, but there exists a number of acceleration methods (e.g., Policy Function Iteration).

4.4.1 An Example: Optimal Growth

Consider a social planner solving:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} = Ak_t^\alpha, \quad k_0 \text{ given,}$$

where $A > 0$, $\alpha \in (0, 1)$, $\beta \in (0, 1)$.

It will be convenient to solve for c_t :

$$\max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln(Ak_t^\alpha - k_{t+1})$$

with k_0 given.

- State variable at $t \rightarrow k_t$ (today's capital)
- Control variable at $t \rightarrow k_{t+1}$ (tomorrow's capital)

Then the Bellman equation is

$$V(k) = \max_{k'} \{ \ln(Ak^\alpha - k') + \beta V(k') \}.$$

Guess the following functional form

$$V^G(k; E, F) = E + F \ln(k),$$

where E and F are unknown constants.

Plug-in $V^G(x; E, F)$ into RHS of Bellman equation

$$\hat{V}(k; E, F) = \max_{k'} \{ \ln(Ak^\alpha - k') + \beta V^G(k'; E, F) \}.$$

First order conditions yield the optimal policy function

$$\frac{\partial}{\partial k'} [\ln(Ak^\alpha - k') + \beta E + \beta F \ln k'] = -\frac{1}{Ak^\alpha - k'} + \frac{\beta F}{k'} = 0 \Rightarrow k'(k) = \frac{\beta F}{1 + \beta F} Ak^\alpha.$$

Substituting into the objective function gives

$$\begin{aligned}\hat{V}(k; E, F) &= \left[\ln\left(\frac{A}{1 + \beta F}\right) + \beta F \ln\left(\frac{\beta F}{1 + \beta F} A\right) + \beta E \right] + \alpha(1 + \beta F) \ln(k) \\ &= E' + F' \ln(k)\end{aligned}$$

$$\Rightarrow \hat{V}(k; E', F') \text{ has the same functional form than } V^G(k; E, F).$$

Finally, pin down the unknown coefficients by solving the system of equations:

$$E^* = \left[\ln\left(\frac{A}{1 + \beta F^*}\right) + \beta F^* \ln\left(\frac{\beta F^*}{1 + \beta F^*} A\right) + \beta E^* \right], \quad F^* = \alpha(1 + \beta F^*).$$

The final solution is

$$V^*(k) = E^* + F^* \ln(k),$$

and the optimal policy function is

$$k'^*(k) = \frac{\beta F^*}{1 + \beta F^*} A k^\alpha.$$

4.5 Stochastic Problems

The previous methodology can easily accommodate uncertainty. A stochastic problem:

$$\max_{\{\mathbf{u}_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t r(\mathbf{x}_t, \mathbf{u}_t) \quad \text{s.t.} \quad \mathbf{x}_{t+1} = g(\mathbf{x}_t, \mathbf{u}_t, \epsilon_{t+1}), \quad \mathbf{x}_0 \text{ given.}$$

ϵ_t is a sequence of iid shocks with cumulative distribution function:

$$\Pr[\epsilon_t \leq e] = F(e).$$

Timing: ϵ_{t+1} is realized at $t + 1$ after $\mathbf{u}_t$ has been chosen. Hence, $\mathbf{x}_t$ is a sufficient statistic at t .

Bellman equation:

$$V(\mathbf{x}) = \max_{\mathbf{u}} \{r(\mathbf{x}, \mathbf{u}) + \beta \mathbb{E}[V(g(\mathbf{x}, \mathbf{u}, \epsilon'))]\}.$$

where

$$\mathbb{E}[V(g(\mathbf{x}, \mathbf{u}, \epsilon'))] = \int V(g(\mathbf{x}, \mathbf{u}, \epsilon')) dF(\epsilon').$$

iid assumption is important for recursive formulation.

If shocks were persistent, instead, the state vector should also incorporate past shock realizations. Previous solution methods still apply.

4.6 Discretisation Methods

In many economic applications, state variables are naturally continuous.

Example 4.6.1. Physical capital, asset holdings, etc.

But when applying computational methods (e.g. VF iteration), we can only evaluate the value function on a finite number of points.

Example 4.6.2. VF iteration:

In the first iteration we generate V^1 by solving

$$V^1(x) = \max_u \{r(x, u) + \beta V^0(g(x, u))\}$$

given an initial guess V^0 . This step requires that we solve the maximisation problem on the RHS for all $x \in \mathcal{X}$. Issue: If $\mathcal{X} \subset \mathbb{R}$, this is not feasible!

How to address this problem? Two approaches:

1. Discretisation of the state space.
2. Polynomial approximation to the value function, or interpolation.

In this course we focus on discretisation methods.

Consider a problem with continuous state and control variables:

$$\mathcal{X} \subset \mathbb{R}, \quad \mathcal{U} \subset \mathbb{R}.$$

Discretisation just replaces $\mathcal{X}$ and $\mathcal{U}$ by finite grids:

$$\hat{\mathcal{X}} = \{x_1, \dots, x_n\}, \quad \hat{\mathcal{U}} = \{u_1, \dots, u_m\}.$$

Now value function becomes a finite list of numbers:

$$\mathbf{V} = \{V_1, \dots, V_n\}$$

This makes the problem much easier to work with!

Back to the optimal growth problem:

$$V(k) = \max_{k'} \{\ln(Ak^\alpha - k') + \beta V(k')\}.$$

In this case $\mathcal{X} = \mathcal{U}$.

If we discretise $\mathcal{X}$, the Bellman equation becomes:

$$V_i = \max_j \{\ln(Ak_i^\alpha - k_j) + \beta V_j\}$$

for all $i = 1, \dots, n$.

To apply Value Function Iteration we iterate on the mapping:

$$V_i^s = \max \left\{ \ln(Ak_i^\alpha - k_1) + \beta V_1^{s-1}, \dots, \ln(Ak_i^\alpha - k_n) + \beta V_n^{s-1} \right\}$$

for all $i = 1, \dots, n$, where s indexes the iteration step.

Note that the maximisation step is much simpler than under the original Bellman equation. This is a key advantage of discretisation methods.

Limits of Discretisation Methods:

- “Curse of dimensionality” in multidimensional state spaces. N points for a one-dimensional state space vs. N^k points for a k -dimensional state space.
- How to choose the grid? Requires some a priori information about the state space, which is sometimes difficult to obtain (e.g. upper and lower bounds).

5 Heterogeneous Agents

We start our journey into macro models with heterogeneous agents, by focusing on household heterogeneity. We use the workhorse model of Aiyagari (1994):

- Households face idiosyncratic shocks to labour productivity.
- Financial markets are incomplete $\rightarrow$ imperfect insurance against idiosyncratic shocks.

5.1 Financial Market Completeness

A key concept in macro models with heterogeneous agents is financial market completeness.

Definition 5.1.1. Financial markets are complete when consumers have access to a sufficiently rich set of assets that allows them to transfer resources across any two dates and/or states of nature.

Essentially, when financial markets are complete, agents are able to insure or “share risk” extensively. Two agents “share risk” if they use state-contingent transfers to increase the expected utility of both, by reducing the risk of at least one. (Ligon 2008)

Consider the following simple model to illustrate the relationship between financial market completeness and insurance. Two periods: $t \in \{0, 1\}$.

- $t = 0$:
 - Consumers trade assets to insure against idiosyncratic shock in $s_1 \in \{s_L, s_H\}$ at $t = 1$.
 - The shock affects income $y(s_1)$ with $y(s_L) < y(s_H)$.
- $t = 1$:
 - Shock is realised.
 - Consumption occurs (no consumption at $t = 0$).

The asset is a claim on consumption at $t = 1$, contingent on realisation of the shock s_1 . Let $b(s_1)$ denote the position on the asset that pays off when the future shock is s_1 , and let $q(s_1)$ be its price. $b(s_1)$ can be positive or negative:

- If $b(s_1) \geq 0$, the consumer buys consumption at $t = 1$ when shock is s_1 .
- If $b(s_1) < 0$, the consumer sells consumption at $t = 1$ when shock is s_1 .

The asset is in zero net supply.

This financial structure is “complete” because consumers can trade assets contingent on future shock realisations.

Budget constraints are as follows:

- At $t = 0$:

$$\sum_{s_1 \in \{s_L, s_H\}} q(s_1) b(s_1) = 0$$

- At $t = 1$:

$$c(s_1) = y(s_1) + b(s_1)$$

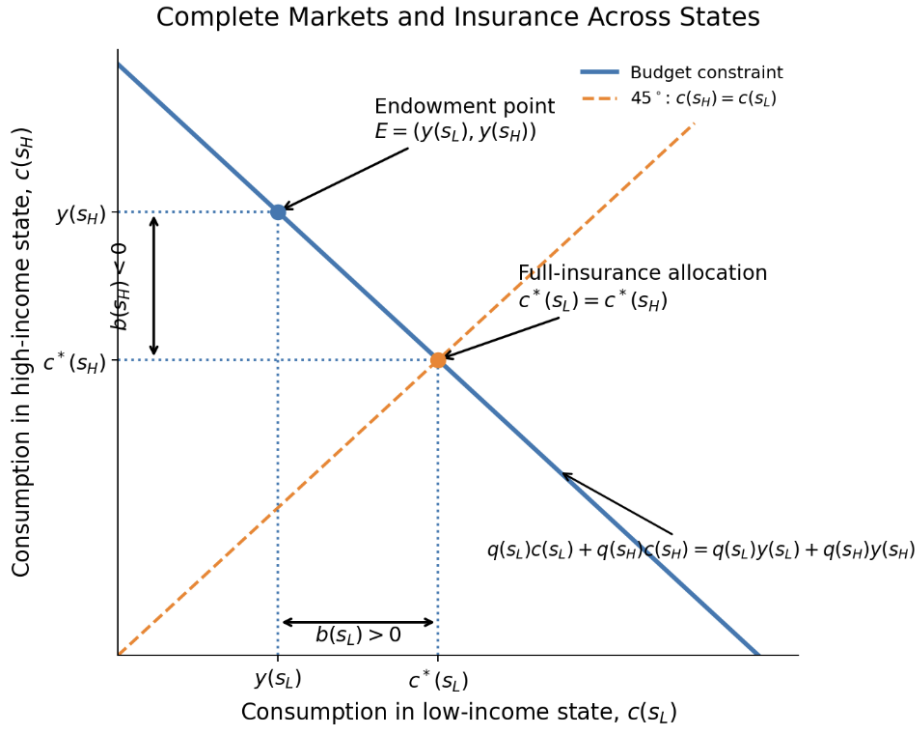


Figure 10: Complete markets: buy claims paying in low-income state and sell claims paying in high-income state, smoothing consumption.

Combining

$$\sum_{s_1 \in \{s_L, s_H\}} q(s_1)c(s_1) = \sum_{s_1 \in \{s_L, s_H\}} q(s_1)y(s_1)$$

Given this asset structure, the consumer achieves insurance, since consumption does not vary one-to-one with the shock realisation.

Now suppose that (for some reason) consumers cannot trade assets contingent on the low shock, i.e., we impose the constraint:

$$b(s_L) = 0$$

This renders financial markets incomplete: agents cannot transfer consumption across states of nature in $t = 1$. It is easy to check that the above implies:

$$c(s_1) = y(s_1) \quad \forall s_1 \in \{s_L, s_H\}$$

Hence, now there is no insurance, as consumption varies one-to-one with the endowment shock.

Suppose all agents are identical *ex ante*. If financial markets are incomplete (so that agents cannot fully insure), idiosyncratic shocks would lead to *ex post* heterogeneity. This is a central mechanism to generate interesting predictions on wealth and income inequality in the Aiyagari model.

5.2 The Aiyagari Model

5.2.1 Households' problem

Discrete time indexed by $t = 0, 1, \dots$

Single consumption good manufactured by a representative firm. Economy populated by a continuum of ex-ante identical households, who are infinitely lived. In each period, households receive idiosyncratic labour productivity shocks

$$s_t \in S = \{\bar{s}_1, \bar{s}_2, \dots, \bar{s}_m\}.$$

where s_t follows a Markov process with transition probability

$$\pi(s'|s) = \text{Prob}(s_{t+1} = s' | s_t = s).$$

Labour income in period t is ws_t , with $w > 0$. Households derive utility from consumption streams $\{c_t\}_{t=0}^\infty$. Expected utility is

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$, u is increasing, twice differentiable, strictly concave, and satisfies the Inada condition $\lim_{c \downarrow 0} u'(c) = +\infty$.

Households can save using two assets: a riskless bond b_t and capital k_t :

- The bond b_t is a risk-free asset, which yields a gross interest rate $(1 + r)$ across periods.

Households save (borrow) at t if $b_{t+1} \geq 0$ (< 0). Households face a borrowing limit denoted by $\underline{b}$. That is:

$$b_{t+1} \geq \underline{b} \quad \forall t.$$

$\underline{b}$ is such that agents never default. The bond is in zero net supply. (i.e., it is an asset that agents trade among themselves.)

- Households can also save by accumulating capital k_t .

Households rent out capital to firms at the rate $\tilde{r}$. Capital holdings of the household evolve according to:

$$k_{t+1} = (1 - \delta)k_t + x_t$$

where $k_t \geq 0$ is the level of capital holdings in period t , $\delta \in (0, 1)$ is the depreciation rate, and x_t is gross investment. Capital is in positive net supply.

Households face the following constraints:

$$c_t + b_{t+1} + x_t = (1 + r)b_t + \tilde{r}k_t + ws_t \quad (\text{Budget Constraint})$$

$$k_{t+1} = (1 - \delta)k_t + x_t \quad (\text{Law of Motion for Capital})$$

Combining we get:

$$c_t + b_{t+1} + k_{t+1} = (1 + r)b_t + (1 + \tilde{r} - \delta)k_t + ws_t \quad (1)$$

Markets for bonds and capital work efficiently. Hence, the following non-arbitrage condition holds:

$$r = \tilde{r} - \delta$$

- LHS: rate of return on bonds
- RHS: rate of return on capital

If $r \neq \tilde{r} - \delta$, the household could make a risk-free profit without costs by rebalancing its portfolio. This won't be consistent with an equilibrium.

Given the non-arbitrage condition $r = \tilde{r} - \delta$, we can write (1) as:

$$c_t + b_{t+1} + k_{t+1} = (1 + r)(b_t + k_t) + ws_t$$

or

$$c_t + a_{t+1} = (1 + r)a_t + ws_t$$

where $a_t \equiv b_t + k_t \geq a$.

As in the Neoclassical Growth Model, we are only able to pin down total asset holdings a_t , not b_t and k_t separately.

Households then solve:

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$\begin{aligned} c_t + a_{t+1} &= (1 + r)a_t + ws_t, \quad \forall(t, s^t), \\ a_{t+1} &\geq a, \quad \forall(t, s^t) \\ a_0, s_0 &\text{ given.} \end{aligned}$$

Households effectively trade only one non state-contingent asset (a_t) to insure against idiosyncratic shocks $s_t \in S = \{\bar{s}_1, \bar{s}_2, \dots, \bar{s}_m\}$. Hence, financial markets are incomplete:

- Agents do not have access to a sufficiently rich set of assets to transfer resources across any two dates and/or states of nature.
- Agents can only achieve imperfect insurance.

Market incompleteness is a key feature of the model. If financial markets were complete, we could recast the model in terms of a representative agent. In that case, heterogeneity “would not matter” in the sense that aggregates would not depend on the distribution of individual quantities.

Intuition:

- Under complete markets, households would fully insure against all future shocks at $t = 0 \dots$ but shocks drive inequality in the model!
- “If the children of Noah had been able and willing to pool risks, (...) then the vast inequality we see today, within and across societies, would not exist.” (Lucas (1992), quoted by Heathcote et al. (2009))

Bellman equation:

$$v(a, s) = \max_{a' \geq a} \left\{ u((1 + r)a + ws - a') + \beta \sum_{s' \in S} \pi(s'|s) v(a', s') \right\}$$

This is an Income Fluctuation Problem: How to allocate consumption across time, given stochastic income fluctuations? Unfortunately, few general/analytical results apply to this class of problems.

Two exceptions:

(1) Precautionary savings motive.

- Precautionary savings are the extra savings caused by future income being stochastic rather than deterministic. HHs build a buffer against the risk of future income drops. Precautionary savings motives are absent under complete markets.
- A sufficient condition for the presence of a precautionary savings motive is that $u''' > 0$ (a.k.a. “prudence”). However, it can also be shown that precautionary savings emerges even in the absence of $u''' > 0$. All we need is that the HH be risk averse and the borrowing limit can bind due to income risk.

(2) Asset accumulation paths based on the value of $\beta(1+r)$.

- $\beta(1+r) < 1$ is a necessary condition for asset accumulation (and consumption) to be bounded. If $\beta(1+r) \geq 1$, asset accumulation (and consumption) grows without bound.

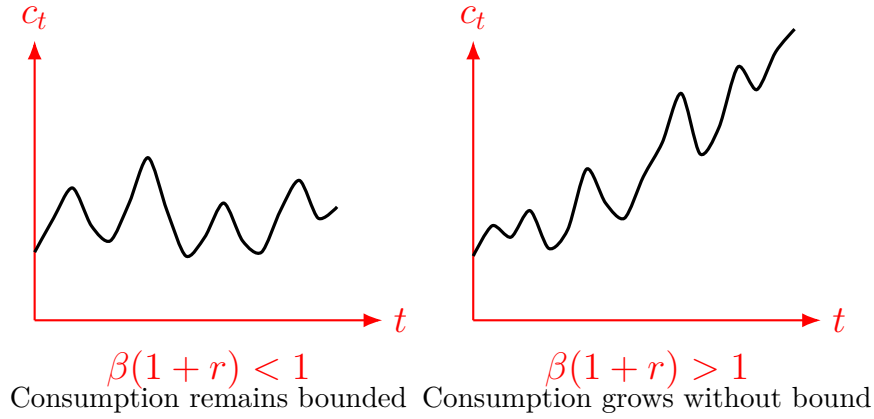


Figure 11: Consumption dynamics under different values of $\beta(1+r)$.

Households are characterized by (a, s) , i.e., their level of wealth and income. Let $\lambda_t(a, s)$ be the unconditional distribution of agents over (a, s) in period t , i.e.

$$\lambda_t(a, s) \equiv \text{Prob}(a_t = a, s_t = s)$$

Households transit across states over time, which induces a law of motion for $\lambda_t(a, s)$.

From now on, we assume that asset holdings are constrained to a grid:

$$a_t \in A = \{\bar{a}_1, \bar{a}_2, \dots, \bar{a}_n\}, \quad \forall t$$

where $\bar{a}_1 = a$ (borrowing limit). This will drastically simplify the derivation of the law of motion for $\lambda_t(a, s)$, and the computation of the equilibrium.

Let $a' = g(a, s) \in A$ be the HH's optimal policy function. This policy function and the Markov transition matrix for s induce the following law of motion for $\lambda_t(a, s)$:

$$\lambda_{t+1}(a', s') = \sum_a \sum_s \lambda_t(a, s) \cdot I(a' = g(a, s)) \cdot \pi(s'|s) \quad (2)$$

where $I(a' = g(a, s)) = 1$ if $a' = g(a, s)$, and $I(a' = g(a, s)) = 0$ otherwise.

Proof. By definition, $\lambda_{t+1}(a', s') = \Pr(a_{t+1} = a', s_{t+1} = s')$. Using the law of total probability, we sum over all possible current states (a, s) :

$$\lambda_{t+1}(a', s') = \sum_{a \in A} \sum_{s \in S} \Pr(a_t = a, s_t = s) \Pr(a_{t+1} = a', s_{t+1} = s' \mid a_t = a, s_t = s).$$

Since $\Pr(a_t = a, s_t = s) = \lambda_t(a, s)$, we get

$$\lambda_{t+1}(a', s') = \sum_{a \in A} \sum_{s \in S} \lambda_t(a, s) \Pr(a_{t+1} = a', s_{t+1} = s' \mid a_t = a, s_t = s).$$

Now decompose the conditional probability:

$$\begin{aligned} & \Pr(a_{t+1} = a', s_{t+1} = s' \mid a_t = a, s_t = s) \\ &= \Pr(a_{t+1} = a' \mid a_t = a, s_t = s) \Pr(s_{t+1} = s' \mid a_{t+1} = a', a_t = a, s_t = s). \end{aligned}$$

The first term is determined by the household policy function:

$$\Pr(a_{t+1} = a' \mid a_t = a, s_t = s) = I(a' = g(a, s)),$$

where

$$I(a' = g(a, s)) = \begin{cases} 1, & \text{if } a' = g(a, s), \\ 0, & \text{otherwise.} \end{cases}$$

The second term is the Markov transition probability for the idiosyncratic shock:

$$\Pr(s_{t+1} = s' \mid a_{t+1} = a', a_t = a, s_t = s) = \Pr(s_{t+1} = s' \mid s_t = s) = \pi(s'|s).$$

So the final law of motion is

$$\lambda_{t+1}(a', s') = \sum_{a \in A} \sum_{s \in S} \lambda_t(a, s) \cdot I(a' = g(a, s)) \cdot \pi(s'|s).$$

This is exactly the law of motion for the distribution in the Aiyagari model.

□

Our goal is to analyze this economy at a steady-state (or “rest point”).

Definition 5.2.1. In steady state, we require that the distribution over states be invariant or stationary. Formally, the stationary distribution $\lambda(a, s)$ is a fixed point of (2):

$$\lambda(a', s') = \sum_a \sum_s \lambda(a, s) \cdot I(a' = g(a, s)) \cdot \pi(s'|s) \quad (*)$$

Clearly, the distribution λ reproduces itself over time (i.e., it is such that $\lambda_t = \lambda_{t+1}$). But there is still “social mobility” even if the distribution is stationary! HHs still move “up and down” the distribution of (a, s) : the rich can become poor, and vice versa.

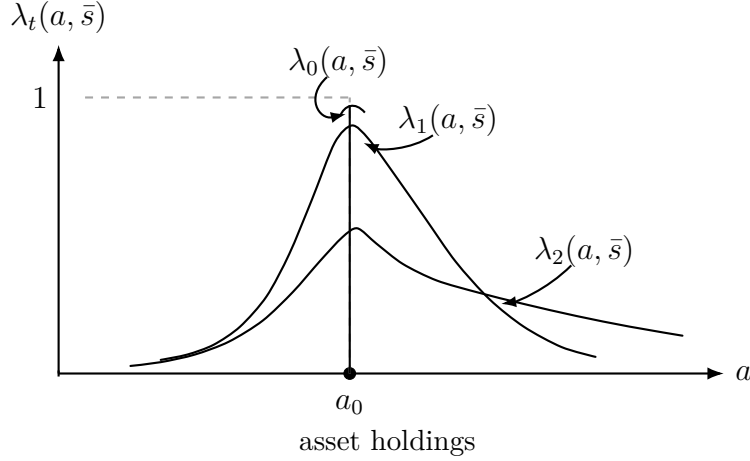


Figure 12: Evolution of the asset distribution conditional on productivity state $\bar{s}$.

A standard approach for computing the distribution function is to start with a guess $\lambda^0(a, s)$, and iterate on the mapping:

$$\lambda^{j+1}(a', s') = \sum_a \sum_s \lambda^j(a, s) \cdot I(a' = g(a, s)) \cdot \pi(s'|s)$$

until convergence (j indexes the iteration step).

This method relies on the fact that the asset space A is discrete. If it weren't, we could discretise A and apply the same method! An alternative method is to simulate asset paths for a large number of households.

5.2.2 Firm's problem

A representative firm produces the consumption good using the Cobb-Douglas technology:

$$F(K_t, N_t) = K_t^\alpha N_t^{1-\alpha}, \quad \text{with } \alpha \in (0, 1),$$

where K_t and N_t denote aggregate capital and labour, respectively. Important: The technology exhibits constant returns to scale.

The firm is owned by the households. The objective of the firm is to maximise profits, by hiring labour and renting capital in spot markets. We assume that markets for capital and labour are perfectly competitive.

The firm thus solves the static problem for all t :

$$\max_{K_t, N_t} F(K_t, N_t) - \tilde{r}K_t - wN_t$$

or

$$\max_{K_t, N_t} F(K_t, N_t) - (r + \delta)K_t - wN_t$$

FOCs give:

$$r = F_K(K_t, N_t) - \delta, \quad w = F_N(K_t, N_t).$$

5.2.3 Stationary Equilibrium

Definition 5.2.2. A stationary equilibrium is a price system (r, w) , a policy function $g(a, s)$, firms' choices K and N , and a distribution over states $\lambda(a, s)$, such that

1. Given (r, w) , the policy function $g(a, s)$ solves the household's problem.
2. Given (r, w) , K and N solve the firm's problem, i.e.,

$$r = F_K(K, N) - \delta \text{ and } w = F_N(K, N).$$

3. The distribution over states $\lambda(a, s)$ is stationary.
4. All markets clear.

This economy has 3 markets:

1. Labour market
2. Asset market
3. Consumption good market

By Walras' law, one of these markets is redundant. We focus on #1-#2.

Let $\Pi(s)$ denote the invariant distribution of the idiosyncratic shock s . $\Pi(s)$ can be computed by iterating on the Markov transition $\pi(s'|s)$. It can be shown that $\Pi(s)$ exists and it is unique under rather weak conditions on $\pi(s'|s)$.

Using $\Pi(s)$, the market clearing condition for labour is:

$$N = \sum_s \Pi(s) s$$

LHS: firm's labour demand at the stationary equilibrium; RHS: total labour supply at the stationary equilibrium.

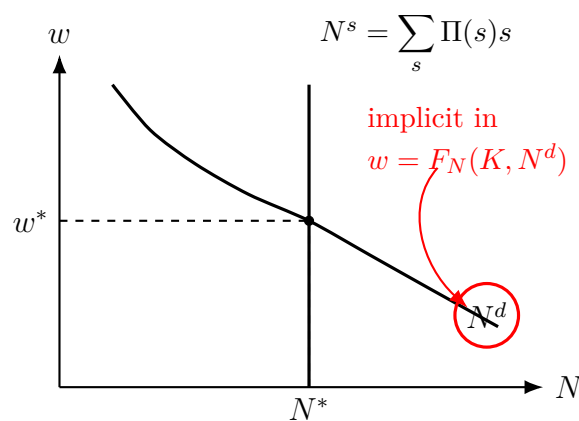


Figure 13: Labor Market Equilibrium

Given K , finding equilibrium in the labour market is trivial:

- Labour supply is exogenous.

- Labour demand is strictly decreasing in the wages, as implicit in $w = F_N(K, N)$.

So given K , we can easily pin down (w, N) in equilibrium. The main challenge is to find the equilibrium level of K .

Recall that we assume that asset holdings are constrained to a grid. Hence, using $\lambda(a, s)$ and $a' = g(a, s)$, market clearing condition for capital is:

$$K = \sum_a \sum_s \lambda(a, s) g(a, s)$$

LHS: firm's capital demand at the stationary equilibrium; RHS: total capital supply at the stationary equilibrium.

We can also write the equilibrium condition in the asset market as

$$K(r) = E_a(r)$$

where

- $K(r)$ is the demand for capital by the firm, as a function of r . By $r = F_K - \delta$, this is a decreasing function of r .
- $E_a(r) \equiv \sum_a \sum_s \lambda(a, s) g(a, s)$ denotes the level of aggregate savings in the economy, given r .

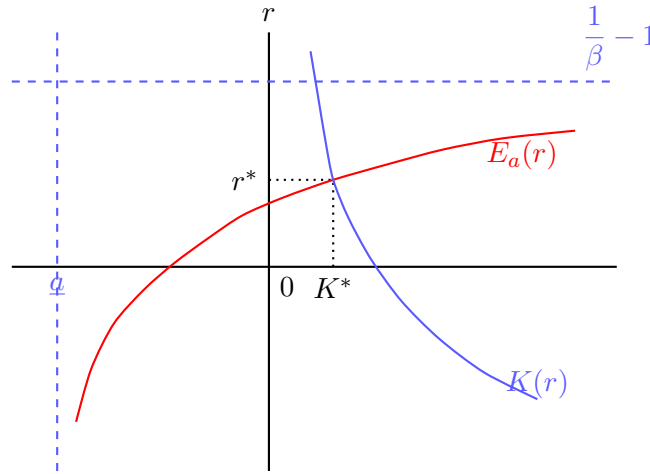


Figure 14: Stationary Equilibrium: Asset Market

Definition 5.2.3 (Revised). A stationary equilibrium is a price system (r, w) , a policy function $g(a, s)$, firms' choices K and N , and a distribution over states $\lambda(a, s)$, such that

1. Given (r, w) , the policy function $g(a, s)$ solves the household's problem.
2. Given (r, w) , K and N solve the firm's problem, i.e.,

$$r = F_K(K, N) - \delta \text{ and } w = F_N(K, N).$$

3. The distribution over states $\lambda(a, s)$ is stationary.

4. All markets clear:

$$N = \sum_s \Pi(s)s$$

$$K = \sum_a \sum_s \lambda(a, s)g(a, s)$$

Computation:

1. Make a guess for $K = K_j$ (j is the iteration step).
2. Compute (w, r) using $r = F_K(K, N) - \delta$ and $w = F_N(K, N)$.
3. Compute the HHs optimal policy function $g(a, s)$.
4. Compute the invariant distribution $\lambda(a, s)$.
5. Compute

$$K_j^* = \sum_a \sum_s \lambda(a, s)g(a, s)$$

6. Check market clearing in the asset market, given tolerance level $\varepsilon > 0$:
 - If $|K_j^* - K_j| > \varepsilon$, update guess to K_{j+1} and go back to #1.
 - If $|K_j^* - K_j| < \varepsilon$, declare victory.

5.3 Applications

5.3.1 Model calibration

In order to solve the model numerically, we need to set the values of the parameters. The idea is to match certain features of interest in the data, which allows us to make quantitative assessments with the models. This exercise usually involves combining two types of “calibration”:

1. External: set of parameter values you can pin down without the need to numerically solve the model (e.g., based on other papers).
2. Internal: set of parameter values which you can only pin down by solving the model numerically.

Suppose the model frequency is one year. **Technology parameters** can be set based on previous studies:

- If we are using Cobb-Douglas production, we set the capital share $\alpha = 1/3$.
- Capital depreciation δ is somewhere between 6–10%.

Preferences:

It is standard to assume that the utility function is CRRA:

$$u(c) = \begin{cases} \frac{c^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \log(c), & \gamma = 1, \end{cases}$$

with $\gamma \in [1, 5]$.

The discount factor β is calibrated to match the wealth-income ratio (K/Y) in the data. In the US, $K/Y \approx 3$.

Labour income process:

The idea is to calibrate the process for the shocks s_t so as to match the dynamics of labour income earnings in the data. Typical source of data for the US is the Panel Study of Income Dynamics (PSID), which is publicly available. The analogue in Australia is the Household, Income and Labour Dynamics in Australia (HILDA). More recently, researchers started to rely more on administrative income data.

Baseline approximation to individual income earnings process is an AR(1):

$$\ln y_t = \rho \ln y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon).$$

Typical estimates yield $\rho \approx 0.95$, $\sigma_\varepsilon = 0.2$.

The estimated AR(1) can then be approximated by a Markov process. A popular way of doing this is Tauchen's method.

Borrowing limit $\underline{a}$: Three standard alternatives:

1. One cheap but common alternative is to set

$$\underline{a} = 0,$$

so that households cannot borrow.

2. On the opposite side of the spectrum, it is common to set

$$\underline{a} = -\frac{\min\{s\}w}{r} \equiv \underline{a}^{nat},$$

where the RHS is known as the “Natural Borrowing Limit.”

This is the maximum amount households can borrow in each period without defaulting under any shock realisation.

3. A middle ground alternative is to calibrate $\underline{a}$ to match certain features of the wealth distribution, for example, % of households with negative net worth.

5.3.2 Quantifying precautionary savings

Recall. Precautionary savings are the extra savings caused by future income being stochastic rather than deterministic.

Aiyagari (1994) quantifies the magnitude of the precautionary savings motive for the US. Idea: Compare level of capital accumulation in model with incomplete markets vs. model with complete markets.

(Huggett (1993) applies similar insights to quantify the effect of imperfect insurance on interest rates in an exchange economy.)

Proposition 5.3.1. *Under complete markets, the equilibrium interest rate is*

$$r^{CM} = \frac{1}{\beta} - 1.$$

Proof. Under complete markets, agents can fully insure against idiosyncratic shocks. Hence, under complete markets the problem that individuals solve is isomorphic to a problem without uncertainty, like this one:

$$\max_{\{c_t, a_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = (1 + r)a_t + \bar{s}, \quad \forall t,$$

where we assumed that $s_t = \bar{s}$ for some $\bar{s}$.

Recursively:

$$V(a) = \max_{a' \geq a} \{u((1 + r)a + \bar{s} - a') + \beta V(a')\}.$$

Assuming an interior solution, the FOC is

$$u'(c) = \beta V'(a'). \tag{1}$$

By the envelope theorem:

$$V'(a) = u'((1 + r)a + \bar{s} - a')(1 + r).$$

Then

$$V'(a) = u'(c)(1 + r) \quad \Rightarrow \quad V'(a') = u'(c')(1 + r).$$

Using this into (1), we get the Euler equation:

$$u'(c) = \beta(1 + r)u'(c').$$

Or using sequential notation:

$$u'(c_t) = \beta(1 + r)u'(c_{t+1}).$$

Recall that there is no aggregate uncertainty in this model. So in a steady state we must have

$$c_t = c_{t+1} \quad \forall t.$$

Then the Euler equation implies that in a steady state under complete markets the interest rate r^{CM} satisfies:

$$1 = \beta(1 + r^{CM}),$$

or

$$r^{CM} = \frac{1}{\beta} - 1.$$

□

The amount of capital accumulated due to precautionary savings is

$$K^* - K^{CM}.$$

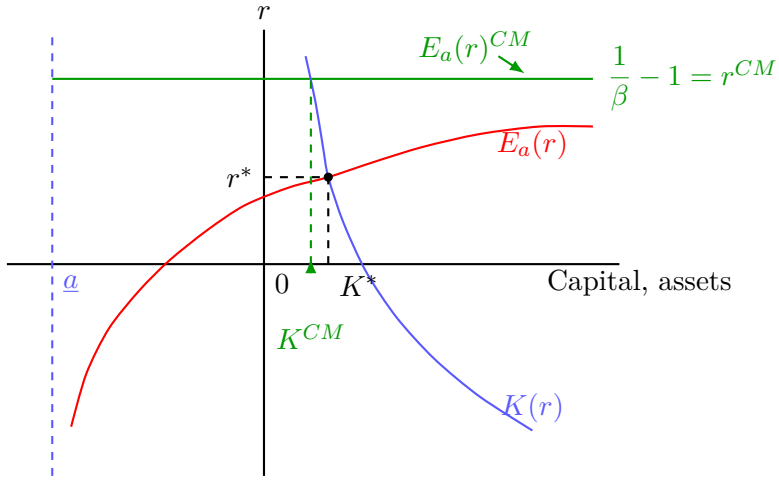


Figure 15: Quantifying Precautionary Savings

We can also work with savings rates $s \equiv S/Y$. Start with the non-arbitrage condition:

$$r = F_K(K, N) - \delta.$$

Using Cobb-Douglas production,

$$F(K, N) = K^\alpha N^{1-\alpha} \Rightarrow F_K(K, N) = \alpha K^{\alpha-1} N^{1-\alpha} = \alpha \frac{Y}{K}.$$

Hence,

$$r = \alpha \frac{Y}{K} - \delta.$$

In steady state, aggregate saving equals investment, and investment is $I = \delta K$. Therefore,

$$s \equiv \frac{S}{Y} = \frac{I}{Y} = \frac{\delta K}{Y}.$$

This implies

$$\frac{Y}{K} = \frac{\delta}{s}.$$

Substituting this into the expression for r , we get

$$r = \frac{\alpha \delta}{s} - \delta.$$

Solving for s ,

$$s = \frac{\alpha \delta}{r + \delta}.$$

- Complete markets savings rate:

$$s^{CM} = \frac{\alpha \delta}{r^{CM} + \delta} = \frac{\alpha \beta \delta}{1 - \beta + \beta \delta}.$$

where α is the capital share, and δ is the depreciation rate.

- Incomplete markets savings rate:

$$s^* = \frac{\alpha\delta}{r^* + \delta},$$

where r^* is the equilibrium interest rate.

Precautionary savings rate:

$$s^* - s^{CM}.$$

Aiyagari quantifies $s^* - s^{CM}$ by calibrating his model to US data. Punchline: The precautionary savings motive is quantitatively not too large.

Some bounds:

1. Log utility & iid earnings shocks $\Rightarrow s^* - s^{CM} \approx 0$.
2. CRRA utility with risk aversion = 5 & very persistent shocks $\Rightarrow s^* - s^{CM} \approx 14$ percentage points.

Intuition: Higher risk aversion and higher persistency of idiosyncratic shocks make households more concerned about self-insurance. The $E_a(r)$ curve shifts right. See graph.

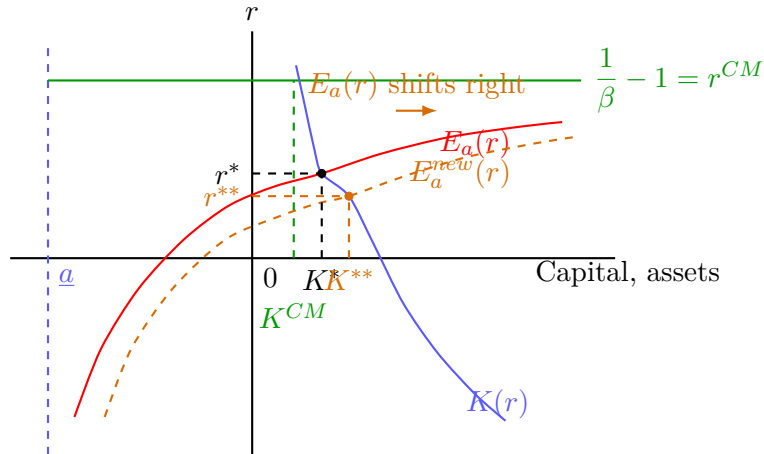


Figure 16: A rightward shift in $E_a(r)$ changes the equilibrium from (K^*, r^*) to (K^{**}, r^{**}) .

5.3.3 Wealth inequality

Aiyagari's model also provides a quantifiable theory of wealth and income distribution. Recall that $\lambda(a, s)$ is non-degenerate!

Question: How much of the observed inequality in wealth and income can be explained by uninsurable idiosyncratic income risk? Aiyagari addresses this question using the model calibrated to US data.

Definition 5.3.2. “Wealth” is households’ net worth, which is a_t in the model.

Definition 5.3.3. “Earnings” is labour earnings (wages $\times$ hours worked in efficiency units), which is $w s_t$ in the model.

Definition 5.3.4. “Income” is revenue from all sources of income before taxes but after transfers, which is $w s_t + r a_t$ in the model.

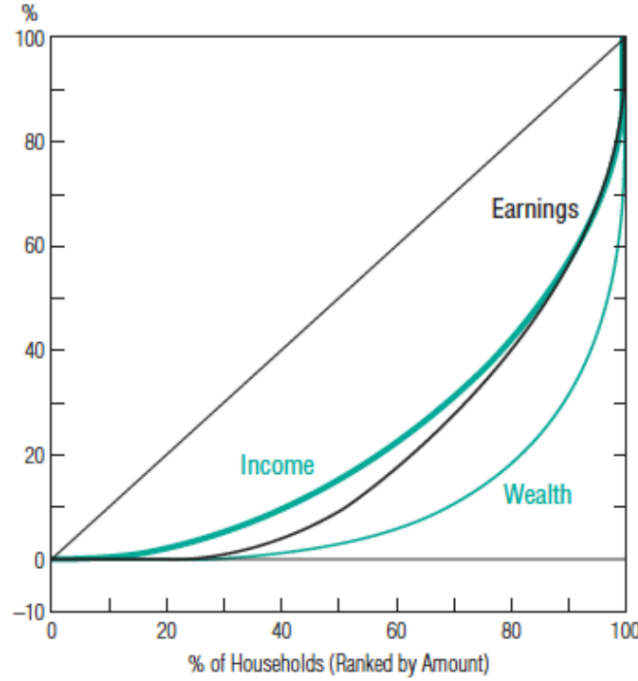


Figure 17: The Lorenz Curves for the U.S. Distributions of Earnings, Income, and Wealth.

We concentrate on the relationship between wealth and earnings. Two key facts based on SCF 1998:

1. Wealth inequality > earnings inequality: Gini of Wealth = 0.80 > 0.61 = Gini of Earnings.

The Gini coefficient here can be understood as the distance between the Lorenz curve and the 45-degree line of perfect equality. If the area between the two lines is called A , and the area below it is called B , then Gini is:

$$Gini = \frac{A}{A + B}.$$

Therefore, the larger the Gini is, the farther the Lorenz curve is from the line of perfect equality, and the more unequal the distribution is.

2. Both distributions are positively skewed, meaning mean > median, but the wealth distribution is more skewed than the earnings distribution: mean/median Wealth = 4.03 > 1.57 = mean/median Earnings.

These facts are robust across time and countries.

	Gini	Bottom 40%	Top 5%	Top 1%
A. U.S. Economy				
Earnings	.63	3.2	31.2	14.8
Wealth	.78	1.7	54.0	29.6
B. Aiyagari (1994)				
Earnings	.10	32.5	7.5	6.8
Wealth	.38	14.9	13.1	3.2

- Qualitatively, the model does a fairly good job matching facts #1-#2 above, except at the top 1%.
- Quantitatively, not so much. In particular: too much wealth at the bottom and too little wealth at the top.

How to fix these?

1. We need to reduce the incentives of the earnings-poor to save. This is generally achieved by introducing means-tested public insurance schemes into the model, such as:
 - Unemployment insurance
 - Food stamps
 - Medical assistance
 - Pensions/social security
 See Hubbard et al. (1995) and others.
2. We need to increase the incentives of the earnings-rich to save. Some modeling solutions:
 - Entrepreneurship (Quadrini (2000), Cagetti and De Nardi (2006))
 - Random and persistent discount factors (Krusell and Smith (1998))
 - Bequests (De Nardi (2004))
 - Changing the income process (Castañeda et al. (2002))

5.3.4 Taxation

Given the absence of complete financial markets, in this model there is a role for the government to provide insurance. Redistributing from the rich to the poor can improve ex-ante welfare.

Suppose that we introduce a government into the model as follows:

The government uses a flat labour income tax τ and a lump-sum transfer t , so that the HHs' BC is

$$c + a' = (1 + r)a + (1 - \tau)ws + t.$$

Question: Which value of τ maximizes ex-ante welfare?

Answer: $\tau = 1$, i.e. a 100% labour income tax. In this way, we can eliminate uncertainty.

Clearly, this is not an interesting framework to analyze optimal policy. In general, redistributive taxation leads to efficiency losses. If τ increases, labour incentives decrease.

To introduce this trade-off between redistribution and efficiency, we need to endogenise labour supply. This model is adapted from Flodén and Lindé (2001). It is based on Violante's lecture notes.

Household's problem is as in Aiyagari, but with the following modifications:

The period utility function is

$$u(c, \ell),$$

where $\ell \in (0, 1)$ is leisure.

Hours worked are $h = 1 - \ell$. Labour earnings are wsh , where w is the wage rate and s is the productivity shock, so sh is effective labour supplied.

Government uses flat labour income tax τ and a lump-sum transfer t to redistribute income. No government debt. No government expenses.

Government's budget:

$$t = \tau wN,$$

where N is the aggregate effective labour supply.

Note: The tax system is progressive in the sense that the average tax increases with individual labour income. Indeed, for an individual with pre-tax labour income wsh ,

$$\text{Average tax} = \frac{\text{net tax paid}}{\text{pre-tax labour income}} = \frac{\tau wsh - t}{wsh} = \tau - \frac{t}{wsh}.$$

As wsh increases, the term $t/(wsh)$ decreases, so the average tax rate increases.

The HHs' problem in recursive form is:

$$v(a, s) = \max_{a' \geq a, h} \left\{ u(c, 1 - h) + \beta \sum_{s' \in S} \pi(s'|s) v(a', s') \right\}$$

subject to

$$c + a' = (1 + r)a + (1 - \tau)wsh + t.$$

Two policy functions:

$$a' = g(a, s), \quad h = h(a, s).$$

HHs can now self-insure against idiosyncratic income shocks in two ways:

- Adjusting asset accumulation
- Adjusting labour supply

Definition 5.3.5. A stationary equilibrium is a price system (r, w) , a tax system (τ, t) , policy functions $g(a, s)$ and $h(a, s)$, firms' choices K and N , and a distribution over states $\lambda(a, s)$, such that

1. Given (r, w, τ, t) , the policy functions $g(a, s)$ and $h(a, s)$ solve the household's problem.
2. Given (r, w) , K and N solve the firm's problem, i.e.,

$$r = F_K(K, N) - \delta \quad \text{and} \quad w = F_N(K, N).$$

3. The distribution over states $\lambda(a, s)$ is stationary.
4. The government's budget constraint is balanced:

$$t = \tau w N.$$

5. All markets clear:

$$N = \sum_a \sum_s \lambda(a, s) s h(a, s) \quad \text{and} \quad K = \sum_a \sum_s \lambda(a, s) g(a, s).$$

Suppose τ is given. Then we can apply the following algorithm:

1. Make a guess for K and N .
2. Compute (w, r) using $r = F_K(K, N) - \delta$ and $w = F_N(K, N)$.
3. Compute t using $t = \tau w N$.
4. Compute the HHs' optimal policy functions $g(a, s)$ and $h(a, s)$.
5. Compute the invariant distribution $\lambda(a, s)$.
6. Check market clearing conditions for assets and labour.
7. Update K and N until convergence.

Now we come back to the question: What is the level of τ that maximizes “social welfare”?

First we need to define “social welfare”. This depends on how we weigh the different types of agents in the model. A standard Social Welfare Function applies equal weights:

$$SWF(\tau) = \sum_a \sum_s \lambda(a, s|\tau) v(a, s|\tau),$$

where $\lambda(a, s|\tau)$ and $v(a, s|\tau)$ are the stationary distribution and value functions, respectively, conditional on τ .

Optimal labour income tax is

$$\tau^* = \arg \max_{\tau} SWF(\tau) = \arg \max_{\tau} \sum_a \sum_s \lambda(a, s|\tau) v(a, s|\tau).$$

τ^* strikes the right balance between insurance and incentives, given the tools that the government has to redistribute.

Flodén and Lindé (2001) compute τ^* for two countries: the US and Sweden. They find:

$$\tau_{US}^* = 0.27 > 0.03 = \tau_{SWE}^*,$$

and welfare gains from implementing τ^* , relative to $\tau = 0$, are larger in the US.

Key differences when calibrating the model for these countries are the income shocks. Income shocks are much less volatile and persistent in Sweden, possibly due to the role of unions. Thus there is more room to improve welfare via public insurance in the US.

Some caveats of Flodén and Lindé's analysis:

1. Other tax instruments?

- Why not use other tax tools?

For example, what about a non-linear labour income tax, capital taxation, or non-linear transfers?

- Life-cycle considerations?

Conesa et al. (2009), for example, address some of these concerns.

2. More general criticism to this class of optimal taxation models: tax and transfer instruments are specified from the onset.

A deeper analysis would be to endogenise the tax tools that the government uses, given the frictions of the environment, e.g. private information on idiosyncratic shocks.

This approach to optimal taxation is a.k.a. the “Mirrlees approach” to optimal taxation.

- Insurance markets, private plus public insurance, are endogenously incomplete.
- Vast literature initiated by Mirrlees (1971). See Mankiw et al. (2009) or Diamond and Saez (2011) for surveys.

6 Heterogeneous Firms

6.1 Model Setup

Infinite horizon, discrete time $t = 0, 1, 2, \dots$. Economy populated by a representative household, and by a continuum of heterogeneous firms.

6.1.1 Firms

Firms differ across productivity s_t and size n_t . Firms make entry, exit, expansion, and contraction decisions based on the realizations of productivity shocks $s_t \in S$. In this lecture, we assume that S is discrete, for simplicity.

Firms’ goal is to maximize the expected discounted present value of their profits. Next we describe the decisions of:

1. Incumbent firms
2. Potential entrants

Incumbent firms produce a homogeneous consumption good by employing labor n_t , according to the technology

$$y_t = f(n_t, s_t),$$

which displays decreasing returns in n . Note: No capital!

s_t follows a Markov process with transition probability $\mathbf{F}(s_{t+1} | s_t)$.

Firms behave competitively in both labor and output markets. Let p_t denote the price of output. Normalize the wage to $w_t = 1$ for all $t \Rightarrow$ labor is the numeraire.

Profits in each period t are given by

$$\pi_t = p_t f(n_t, s_t) - n_t - p_t c_f - g(n_t, n_{t-1}).$$

The term c_f is a fixed operating cost, which gives rise to firm exit. The function $g(n_t, n_{t-1})$ encapsulates labor adjustment costs.

- From now on, we assume:

$$g(n_t, n_{t-1}) = \tau \cdot \max(0, n_{t-1} - n_t),$$

where τ is a tax of labor destruction.

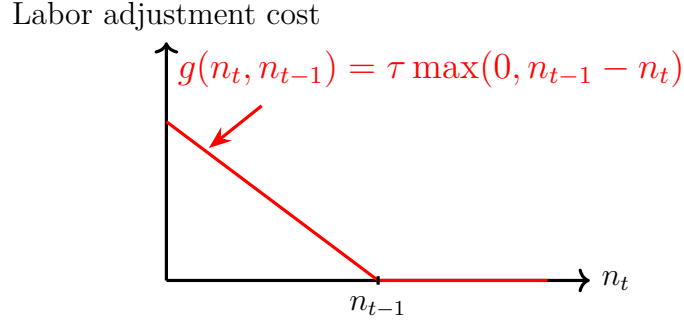


Figure 18: Labor adjustment cost as a function of current employment

- If a firm exits, it must pay the adjustment cost $g(0, n_{t-1})$.

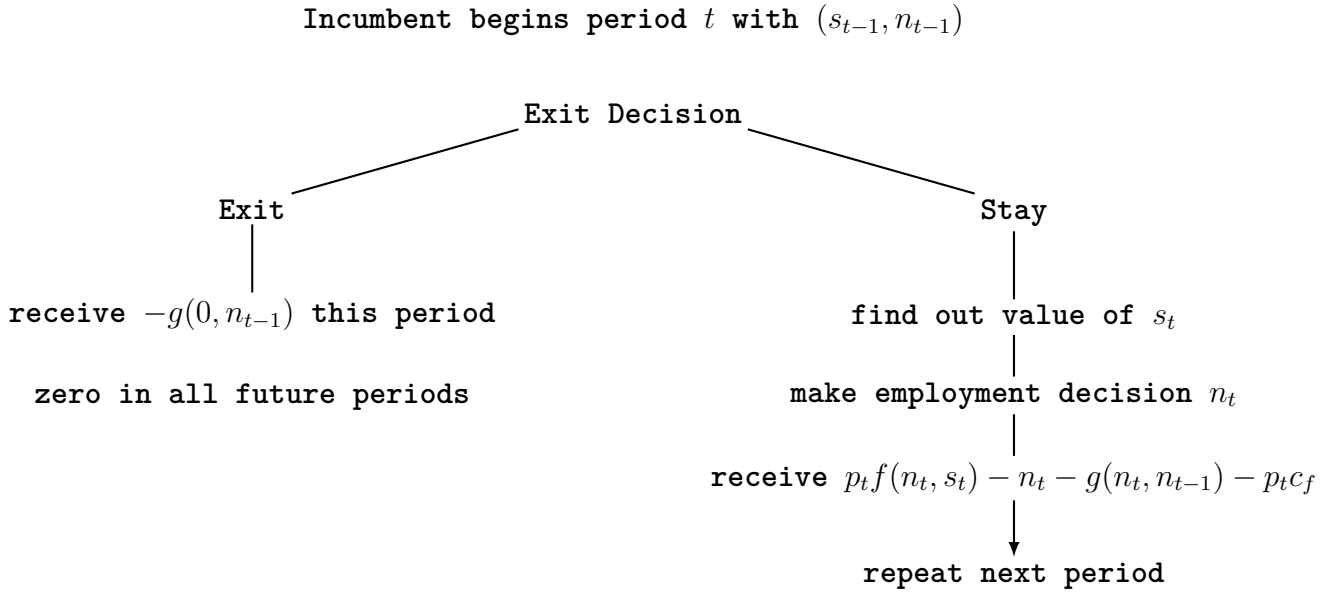


Figure 19: Timing of incumbent firms' decisions

The problem of an incumbent can be written recursively as

$$W(s, n; p) = \max_{n' \geq 0} \{p f(n', s) - n' - p c_f - g(n', n) + \beta \max [EW(s', n'; p), -g(0, n')]\}.$$

Note:

- Individual states are (s, n) , p is a parameter.

- $\beta \in (0, 1)$ is the household's discount factor.

This Bellman equation produces two optimal decision rules:

- $N(s, n; p)$: optimal choice of current employment.
- $X(s, n; p) \in \{0, 1\}$: optimal stay/exit choice, equal to 0 if stay.

In each period, there is a continuum of ex ante identical potential entrants. Entrants incur a one-time entry cost c_e in terms of output.

Upon paying entry cost, new entrants draw initial productivity s from distribution ν . Initial productivity draw is iid across entering firms, across periods, and across the number of entering firms.

After initial period, entering firms behave as incumbent firms presented earlier.

Let

$$W^e(p) = \sum_s W(s, 0; p) \nu(s),$$

where W is the incumbent's value function defined above.

Potential entrants then solve

$$\max_{E \in \{0, 1\}} E \cdot [W^e(p) - pc_e]$$

where $E = 1$ corresponds to entry.

Question: What is the main source of firm heterogeneity in this model?

Answer: Firm productivity shocks coupled with decreasing returns.

Intuition: Assume no adjustment costs, no fixed costs, no entry, and $f(n, s) = sn^\theta$ with $\theta \in (0, 1)$. Then in each period the firm solves

$$\max_n psn^\theta - n \Rightarrow n = (p\theta s)^{\frac{1}{1-\theta}}.$$

- Heterogeneity in s leads to heterogeneity in n .
- Scale is finite and determined, which leads to a well-defined distribution of firm sizes.

Question: What would happen under constant returns and increasing returns?

Answer. Under constant returns to scale, firm size would not be well determined. For example, suppose that $f(n, s) = sn$. Then the firm solves

$$\max_n psn - n = \max_n (ps - 1)n.$$

If $ps > 1$, the firm wants to hire an infinite amount of labor. If $ps < 1$, the firm hires zero labor. If $ps = 1$, the firm is indifferent over any level of employment. Therefore, productivity heterogeneity would not generate a well-defined distribution of firm sizes.

Under increasing returns to scale, larger firms become more productive as they expand. This creates a tendency for firms to grow without bound and may lead to market concentration or monopoly. Hence, increasing returns are inconsistent with the competitive firm-size distribution that the Hopenhayn–Rogerson model is designed to study.

Key difference between the two models. The underlying source of agent heterogeneity is fundamentally different from Aiyagari (1994). In Aiyagari, households are heterogeneous due to

idiosyncratic risk plus incomplete financial markets, a financial friction. In Hopenhayn-Rogerson, firms are heterogeneous ultimately due to idiosyncratic risk plus decreasing returns, a technological feature.

6.1.2 Government and Household

The government's role is almost trivial, but we define it for completeness. It collects revenues from taxing firms for job destruction at the rate τ . **Recall.**

$$g(n_t, n_{t-1}) = \tau \cdot \max(0, n_{t-1} - n_t).$$

Such revenues are rebated in a lump-sum fashion to the representative household.

We implicitly assume that individuals can fully diversify idiosyncratic risk. Hence, we can work with a representative individual/household. Preferences:

$$\sum_{t=0}^{\infty} \beta^t [\log(c_t) - AN_t],$$

where $A > 0$, c_t is consumption, and N_t is labor supply.

The household owns the firms and collects profits/dividends in every period.

Recall, in steady state, $\beta(1+r) = 1$. This implies that aggregate consumption is constant, and that individual choices of c and N solve the static problem:

$$\max_{c, N} \log(c) - AN$$

subject to

$$pc \leq N + \Pi + R,$$

where Π denotes firms' profits and R are government transfers.

This problem yields the labour supply function:

$$L^s(p, \Pi + R) = \frac{1}{A} - (\Pi + R).$$

6.2 Stationary Equilibrium

Incumbents are indexed by (s, n) . Hence, we can define the distribution over incumbents in period t by

$$\mu_t(s, n) \equiv \text{Prob}(s_t = s, n_t = n).$$

Similarly, let $M_t \geq 0$ denote the mass of new entrants in period t .

Note: New entrants are ex ante identical. Hence, the mass of new entrants is given by a scalar rather than a distribution.

In steady state, the measure of incumbents and new entrants are time invariant, say $\mu(s, n)$ and M . We need these steady state measures to compute aggregates and find the steady state equilibrium!

How to compute μ at the steady state? For a given M , law of motion for μ_t is given by:

$$\begin{aligned} \mu_{t+1}(s', n') &= \sum_s \sum_n \mu_t(s, n) \cdot I(n' = N(s, n; p)) \cdot I(X(s, n; p) = 0) \cdot \mathbf{F}(s'|s) \\ &\quad + M_t \cdot \nu(s') \cdot I(n' = 0) \end{aligned}$$

Intuition:

- First line on RHS: Contribution of incumbents.
- Second line on RHS: Contribution of new entrants.

We can write the law of motion in (1) more succinctly as

$$\boldsymbol{\mu}_{t+1} = \mathbf{T}(\boldsymbol{\mu}_t, M_t, p),$$

where $\mathbf{T}$ is a transition operator.

At steady state:

$$\boldsymbol{\mu} = \mathbf{T}(\boldsymbol{\mu}, M, p).$$

Given $(\boldsymbol{\mu}, M)$, we can compute all aggregate variables. In particular:

- Labor demand:

$$L^d(\boldsymbol{\mu}, M, p) = \sum_s \sum_n N(s, n; p) \boldsymbol{\mu}(s, n) + M \sum_s N(s, 0; p) \boldsymbol{\nu}(s).$$

Intuitively, labor demand = incumbents + new entrants.

- Labor supply:

$$L^s(p, \Pi(\boldsymbol{\mu}, M, p) + R(\boldsymbol{\mu}, M, p)) = \frac{1}{A} - (\Pi(\boldsymbol{\mu}, M, p) + R(\boldsymbol{\mu}, M, p)),$$

where $\Pi(\boldsymbol{\mu}, M, p)$ and $R(\boldsymbol{\mu}, M, p)$ are profits and government transfers consistent with $(\boldsymbol{\mu}, M, p)$.

Definition 6.2.1 (Stationary Equilibrium). A stationary equilibrium consists of a measure of incumbents $\boldsymbol{\mu}^*$, a mass of entrants M^* , and an output price p^* such that three conditions hold:

1. Free entry:

$$W^e(p^*) = p^* c_e.$$

2. The probability distribution $\boldsymbol{\mu}^*$ is stationary, i.e.,

$$\boldsymbol{\mu}^* = \mathbf{T}(\boldsymbol{\mu}^*, M^*, p^*).$$

3. Labor market clears:

$$L^d(\boldsymbol{\mu}^*, M^*, p^*) = L^s(p^*, \Pi(\boldsymbol{\mu}^*, M^*, p^*) + R(\boldsymbol{\mu}^*, M^*, p^*)).$$

Computation sketch

1. Find the p^* that satisfies the free entry condition. To this end, we need to solve the Bellman equation $W(s, n; p)$ for various values of p , until the free entry condition holds.
2. Given p^* , solve for $\boldsymbol{\mu}^*$ and M^* such that

$$\boldsymbol{\mu}^* = \mathbf{T}(\boldsymbol{\mu}^*, M^*, p^*)$$

and

$$L^d(\boldsymbol{\mu}^*, M^*, p^*) = L^s(p^*, \Pi(\boldsymbol{\mu}^*, M^*, p^*) + R(\boldsymbol{\mu}^*, M^*, p^*)).$$

6.3 Quantitative Results

- Parameterization:

$$\text{Technology} \rightarrow f(n, s) = sn^\theta, \quad \theta \in (0, 1)$$

$$\text{Firm productivity} \rightarrow \log(s_t) = a + \rho \log(s_{t-1}) + \epsilon_t, \quad \epsilon \sim N(0, \sigma_\epsilon^2)$$

- Externally calibrated parameters:
 - $\theta = 0.64$, based on labor share.
 - $\beta = 0.8$, based on 1 period = 5 years.
- Other parameters are calibrated to match US data moments, assuming that US economy has $\tau = 0$.

Effect of Changes in τ (Benchmark Model)

	$\tau = 0$	$\tau = .1$	$\tau = .2$
Price	1.00	1.026	1.048
Consumption (output)	100	97.5	95.4
Average productivity	100	99.2	97.9
Total employment	100	98.3	97.5
Utility-adjusted consumption	100	98.7	97.2
Average firm size	61.2	61.8	65.1
Layoff costs/wage bill	0	.026	.044
Job turnover rate	.30	.26	.22
Serial correlation in $\log(n)$	.92	.94	.94
Variance in growth rates	.55	.45	.39

$\tau = .1$ ($\tau = .2$) correspond to 6 months' (1 year's) wages.

As τ increases, firms become more cautious in adjusting n , so n becomes more persistent, and variance of n falls. Job destruction decreases, as reflected by lower turnover. But total employment and welfare fall quite a bit.

Effect of τ on Decision Rules

$\log s$	$\tau = .1$		$\tau = .2$	
	n_l	n_u	n_l	n_u
1.83	1.36	1.78	1.18	1.98
4.75	21.7	26.7	21.0	32.8
10.5	194	238	181	282
19.9	1,110	1,410	1,036	1,617
27.3	2,610	3,316	2,522	3,935

Inefficiencies are also reflected by inaction regions which widen significantly with τ .

Absolute Deviations from $MPL = 1/p$

Size of Deviation (%)	Fraction of Firms within Interval	
	$\tau = .1$	$\tau = .2$
0–3	.30	.00
3–5	.45	.12
5–10	.15	.78
10–15	.00	.05
> 15	.00	.05

Last way to see effect of policy distortions is via differences between MPL and $1/p$, real wage. Size of deviations increase for most firms.